Shadows, Fibres and Secants of the Posterior Tower

Conditional expectations, Beck defects, and positive tropicalisation

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Abstract

The three-level posterior tower [1] has two candidate mixed distributive laws, and both fail some Beck axioms. This paper studies the failures through two operations on the tower's levels: *pooling*, which replaces a law by conditional expectations of it, and *tropicalisation*, which replaces probabilities by their large-deviation rates. It unifies and corrects three earlier manuscripts, on the Giry-monad formulation, on conditional shadows, and on fibres and secants. The changes are listed in appendix A.

Algebra. For finite partitions A, B of a full-support probability space, five conditions are equivalent:

- $E_A E_B = E_B E_A$;
- $E_A E_B = E_{A \wedge B}$;
- A and B are conditionally independent given the meet $A \wedge B$;
- every connected component of the bipartite cell graph has a rank-one contingency block;
- every singular value of the normalised contingency matrix is 0 or 1.

The identity $E_A E_B = E_{A \vee B}$ holds only when $A = B$. In general $(E_A E_B)^n \to E_{A \wedge B}$ at the exact rate c^{2n-1}, so interchange holds laxly, through the meet.

Shadows. A coherent context determines a star-shaped probability space centred at the level-one predictive. Shadows are conditional expectations on this fixed space. They form a commutative band and satisfy interchange strictly. Shadows defined by label maps on laws do not compose as conditional expectations.

Topology and order. The barycentre has two continuous splittings, $e_\perp = \eta\mu$ and $e_\top = \mathsf{D}(\eta)\mu$. These form a left-zero band and bound every barycentre fibre in convex order. A coherent context is a three-step simplex-valued martingale. The shift law's level-one unit defect is the full convex-order diameter of a fibre; its multiplication defect is one conditional-expectation step.

Categorical structure. We construct the lifting functor from contexts to measure spaces in two versions and prove naturality of the shadows along measure-preserving maps. The double category is the quintet construction of a locally preordered 2-category, in which interchange is automatic. In this language each Beck defect computed so far is a single conditional-expectation step.

Geometry. Over a real closed non-Archimedean field the valuation is a monad morphism to the min-plus distribution monad. Classical barycentre fibres map onto tropical fibres. Positive mixtures map exactly onto tropical secants. The valuation factors through the Berkovich analytification, and on positive points the square commutes. The Bayes posterior tropicalises to the hard assignment, so the Voronoi walls of nearest-atom pooling are tropical walls. We also correct the Gaussian fusion square, the finite stochastic-to-convex functor and the moment-variety programme.

Cohomology. Conditional expectations are the restriction maps of the L^2 presheaf on the lattice of σ-algebras. Its Beck–Chevalley condition is exactly conditional independence given the meet, with defect norm c. Compatible sections always glue in finite dimensions, and gluing fails only when $c = 1$. The Beck comparisons are restriction maps, never isomorphisms. Tropical fibres are covered by assignment cells. The nerve computes their cohomology, and Farkas certificates of impossible assignment patterns, the tropical form of the tower's lifting certificates, determine it. For Hardy–Weinberg mixtures the fibre has $b_0 = 2$ off the collision locus and $b_0 = 1$ on it, with $b_1 = 0$, so the collision wall is a jump in H^0.

Hallucination. In a three-token model observed through pairwise windows, a hallucination (an obstructed claim) is a failure of gluing. It occurs below a straight wall in source-weight space, at agreement $\frac{1}{3}$, and is certified by a Bell-type functional. Above the wall, the two overlapping windows satisfy Beck–Chevalley iff the tokens are Markov through the shared token. The defect is $c = |2\varepsilon - 1|/(2\varepsilon)$ at the symmetric section. Tropicalisation sees only support-level hallucinations. A pre-registered test on public GPT-3.5 and GPT-4 answers finds that cross-run disagreement detects wrong answers in all five tasks (AUROC 0.69–0.88), and that the window defect exceeds its Markov null in three of them. One worked example, the Hardy–Weinberg atlas, illustrates every layer in seven figures generated by a script. Four scripts check every finite claim, with 30, 8, 10 and 9 checks, all passing.

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1 Introduction

1.1 The problem

A three-level posterior tower records three things: a predictive $\rho \in \mathsf{D}X$, a law $\Pi \in \mathsf{D}^2X$ over predictives, and a law $\Omega \in \mathsf{D}^3X$ over such laws. It also carries a shadow Σ of labelled hypotheses. A candidate mixed distributive law on the tower may fail some Beck axioms, and the failures are comparisons between constructions, not merely numerical discrepancies. The motivating question is whether two ways of pooling information organise these comparisons, and whether the comparisons have a geometric meaning. Three logically separate tasks are involved:

(1) define the pooling operators (shadows) and prove their operator identities;

(2) build a category or double category in which the Beck comparisons are well-typed cells;

(3) locate the comparisons geometrically: in barycentre fibres, in convex order, and after tropicalisation, in secant geometry.

None of these implies another, and the paper keeps them apart.

1.2 Guiding principles

(i) *Compose operators on a fixed probability space.* Composing measure-valued maps that recompute their partitions after each step is a different operation. Most earlier difficulties came from mixing the two.

(ii) *Conditional expectations interact through the meet, never the join.*

(iii) *Probability tropicalises cleanly because it is positive.* Leading terms of positive series cannot cancel, so the valuation is a semiring homomorphism. Every geometric statement in the paper rests on this.

1.3 Main results

(A) **Commutation** (theorem 3.3): five equivalent conditions, a spectral and a combinatorial criterion among them. The join identity holds only for equal σ-algebras (proposition 3.4).

(B) **Rate and lax interchange** (theorem 3.6 and corollary 3.8).

(C) **Shadow algebra** (theorem 4.9): the canonical shadows of a context form a commutative band.

(D) **Fibres and martingale** (propositions 5.1 and 5.2 and theorem 5.3).

(E) **Beck defects located** (propositions 5.4 and 7.2).

(F) **Lifting functor and naturality** (propositions 4.6, 6.5 and 6.6); quintet double category (proposition 6.10).

(G) **Tropical tower** (theorem 11.3), **fibre lifting** (theorem 11.6), **tropical secants** (theorem 11.9) and the **Berkovich square** (theorem 11.11).

(H) **Zero-temperature shadows and walls** (propositions 12.1 and 12.2).

(I) **Cohomology**: Beck–Chevalley for the L^2 presheaf is conditional independence (theorem 14.4); gluing and the degree-one obstruction (theorem 14.6); fibre cohomology from the nerve and its certificates (theorem 15.3); the Hardy–Weinberg computation (theorem 15.5).

(J) **Hallucination**: a three-token model in which hallucination is a gluing failure with a certificate (proposition 16.1), and the window shadows' Beck–Chevalley defect measures non-Markov coupling (proposition 16.2); its tropical shadow is support-level only (proposition 16.3).

1.4 Status

Statement	Status
operator commutation, rate, lax interchange	proved, finite and general
canonical shadows, band, naturality	proved
fibres, martingale, location of the (U) and (M) defects	proved
lifting functor, quintet double category	proved
a 2-category in which D, C_3 and the laws act monotonically	open (open problem 6.11)
tropical tower, fibre lifting, tropical secants, Berkovich square	proved
tropical identifiability inside a model	open (open problem 12.5)
wall-crossing formula for Gaussian mixture towers	proposed (conjecture 12.4)
L^2 presheaf: Beck–Chevalley, gluing, comparisons are restrictions	proved
fibre cohomology via assignment-cell nerves; certificates determine it	proved; Hardy–Weinberg computed
lifting obstructions inside a model as cohomology classes	open (open problem 12.7)
three-token hallucination: gluing failure, window Beck–Chevalley, tropical shadow	proved
window defect on real model outputs (pre-registered Study 1)	nonzero in 3 of 5 tasks
obstruction below $\frac{1}{3}$ in pairwise-elicited model answers	kit ready (Study 3); needs model access

1.5 Conventions

X and B denote finite sets, and D is the finite distribution monad with $\eta_X(x) = \delta_x$ and $\mu_X(\Pi) = \sum_\rho \Pi(\rho)\rho$. We identify $\mathsf{D}B$ with the simplex $\Delta_B \subset \mathbb{R}^B$, with the ℓ^1 metric. A *finite probability space* (I, p) has $p(i) > 0$ for all i, and its σ-algebras are partitions. For a partition A, E_A is conditional expectation, acting coordinatewise on $\mathbb{R}^X$-valued functions: $(E_A Z)(i) = \sum_{j \in A(i)} p(j)Z(j)/p(A(i))$, where $A(i)$ is the cell of i. In the tower paper [1], the letters L and R also denote the two sides of the (Cm) diagram. Here L_c and R_c always denote shadows (definition 4.8).

2 The posterior tower

2.1 Levels and contexts

For a finite set B, the iterated distribution spaces $\mathsf{D}B, \mathsf{D}^2B, \mathsf{D}^3B, \ldots$ carry barycentre maps satisfying the monad laws. A hypothesis is $h = (\theta, r) \in \mathsf{Hyp}B = \Theta \times \mathsf{D}B$, with predictive $q(h) = r$. Hyp is a functor and q is natural [1, Def. 3.1]. The shadow space is $\mathsf{Sh}(B) = \mathsf{D}(\mathsf{Hyp}B)$. A context is $c = (\rho, \Pi, \Omega, \Sigma)$ with $\rho \in \mathsf{D}B$, $\Pi \in \mathsf{D}^2B$, $\Omega \in \mathsf{D}^3B$ and $\Sigma \in \mathsf{Sh}B$.

Definition 2.1 (Coherent context)**.** c is *coherent* if

$$\mu(\Pi) = \rho, \qquad \mu(\Omega) = \Pi, \qquad \mathsf{D}(q)(\Sigma) = \Pi. \tag{1}$$

Remark 2.2 (Typing)**.** The third condition is meaningful because q lands in $\mathsf{D}B$, the underlying space of Π. Any variant of the shadow component must come with its own projection to $\mathsf{D}B$ and its action on maps $f\colon B \to B'$. For $\mathsf{Hyp} = \Theta \times \mathsf{D}(-)$ this action is $\mathsf{Hyp}(f)(\theta, r) = (\theta, f_* r)$.

2.2 The absolute tower and the two laws

The absolute tower $\mathsf{C}_3X = X \times \mathsf{D}X \times \mathsf{D}^2X \times \mathsf{D}^3X \times \mathsf{Sh}X$ is a comonad, with comultiplication $\delta(x, c) = ((x, c), \mathsf{E}_3(j_c)(c))$, where $j_c(y) = (y, c)$ [1, §6]. Applying it at $\mathsf{D}X$ gives $\mathsf{C}_3(\mathsf{D}X) = \mathsf{D}X \times \mathsf{D}^2X \times \mathsf{D}^3X \times \mathsf{D}^4X \times \mathsf{Sh}(\mathsf{D}X)$. For a sample $M = \sum_i a_i \delta_{(x_i, c_i)}$:

- the *shift law* λ^σ keeps each sample's context entries, moving them one level up;
- the *averaging law* $\bar{\lambda}$ pushes contexts through the unit and averages them.

2.3 Beck axioms

A mixed distributive law $\lambda\colon \mathsf{DC} \Rightarrow \mathsf{CD}$ must satisfy the unit (U), counit (Co), multiplication (M) and comultiplication (Cm) diagrams [1, Def. 2.1]. The shift law satisfies (Co) and fails (U), (M) and (Cm). The averaging law satisfies (U), (Co) and (M) and fails (Cm) [1, Thms. 8.3, 8.6, 8.7]. The base-relative tower has a strict law [1, Thm. 9.8]. A failure produces two parallel composites with the same source and target, and the comparison between them is the *Beck defect*.

3 Algebra of conditional expectations

3.1 The partition lattice

Partitions of I form a lattice under refinement. The *join* $A \vee B$ has cells $a \cap b$. The *meet* $A \wedge B$ is the finest common coarsening. As σ-algebras, $A \vee B = \sigma(A \cup B)$ and $A \wedge B = A \cap B$.

Lemma 3.1 (The meet is π_0 of a bipartite graph)**.** *Let $G(A, B)$ be the bipartite graph on the cells of A and of B, with an edge a–b whenever $a \cap b \neq \emptyset$. The cells of $A \wedge B$ are the unions over the connected components of $G(A, B)$, and $L^2(A) \cap L^2(B) = L^2(A \wedge B)$.*

Proof. A set lies in $A \cap B$ iff it is a union of A-cells and a union of B-cells, i.e. iff it is closed under the edge relation. A function lies in $L^2(A) \cap L^2(B)$ iff it is constant on cells of both partitions, i.e. iff it is constant on components. □

3.2 The commutation theorem

Write $N_{ab} = p(a \cap b)$ for the contingency matrix and $K = D_A^{-1/2} N D_B^{-1/2}$, with $D_A = \mathrm{diag}(p(a))$ and $D_B = \mathrm{diag}(p(b))$.

Lemma 3.2 (A sufficient pair)**.** *On any probability space, let $C \subseteq A \cap B$ be sub-σ-algebras such that A and B are conditionally independent given C. Then $E_A E_B = E_C = E_B E_A$, and $C = A \cap B$ up to null sets.*

Proof. For $V \in L^2(B)$ and $U \in L^2(A)$, conditional independence gives $\mathbb{E}[UV] = \mathbb{E}[\mathbb{E}[U \mid C]\mathbb{E}[V \mid C]] = \mathbb{E}[U\,\mathbb{E}[V \mid C]]$, using $C \subseteq A$. So $E_A V = E_C V$, and hence $E_A E_B = E_C E_B = E_C$, using $C \subseteq B$. The other order is symmetric. The range of E_C is $L^2(C)$, and that of the commuting product is $L^2(A) \cap L^2(B) = L^2(A \cap B)$. □

Theorem 3.3 (Commutation)**.** *For partitions A, B of a finite probability space the following are equivalent:*

(i) $E_A E_B = E_B E_A$*;*

(ii) $E_A E_B = E_{A \wedge B}$*;*

(iii) *A and B are conditionally independent given $A \wedge B$;*

(iv) *on each component γ of $G(A, B)$, the block $N|_\gamma$ has rank one, i.e. $p(a \cap b)p(\gamma) = p(a)p(b)$ for all cells $a, b \subseteq \gamma$;*

(v) *every singular value of K is 0 or 1.*

The equivalences (i)⇔(ii)⇔(iii) hold on every probability space, with $A \wedge B$ read as $A \cap B$. The value 1 in (v) has multiplicity equal to the number of components. Nestedness ($A \subseteq B$ or $B \subseteq A$) implies (iv).

Proof. (iii)⇒(ii) is lemma 3.2. (ii)⇒(i): $E_{A \wedge B}$ is self-adjoint, so $E_B E_A = (E_A E_B)^* = E_A E_B$. (i)⇒(ii): the product of commuting orthogonal projections is the projection onto the intersection of their ranges, which is $L^2(A \wedge B)$ by lemma 3.1. (ii)⇒(iii): with $C = A \wedge B$, $\mathbb{E}[UV \mid C] = \mathbb{E}[U\,E_A V \mid C] = \mathbb{E}[U\,E_C V \mid C] = E_C U \cdot E_C V$, since $E_A V = E_A E_B V = E_C V$ is C-measurable.

(iii)⇔(iv): on a finite space it suffices to test indicators. Cells lie inside single components, so conditional independence given C reads $p(a \cap b \mid \gamma) = p(a \mid \gamma)p(b \mid \gamma)$. This says $N|_\gamma$ is the outer product of its margins divided by its total, which for a nonnegative matrix is the same as rank one.

(ii)⇔(v): in the orthonormal bases $1_a/\sqrt{p(a)}$ and $1_b/\sqrt{p(b)}$, $E_A|_{L^2(B)}$ has matrix K, whose singular values are the cosines of the principal angles between $L^2(A)$ and $L^2(B)$. The value 1 occurs exactly on $L^2(A \wedge B)$. Condition (ii) holds iff all other angles are $\pi/2$. If $A \subseteq B$, every A-cell is a component and (iv) is trivial. □

Check A1 confirms the five-way equivalence on 600 random instances, 449 commuting and 151 not.

Proposition 3.4 (The join identity)**.** *$E_A E_B = E_{A\vee B}$ holds iff $A = B$ up to null sets.*

Proof. The range of $E_A E_B$ lies in $L^2(A)$. If it equals $L^2(A \vee B) \supseteq L^2(B)$, then $B \subseteq A$, so $E_A E_B = E_B$ and $E_{A\vee B} = E_A$, whence $A = B$. □

Example 3.5 (A non-commuting pair)**.** Let $I = \{1,2,3,4\}$ with $p = (\frac{3}{8}, \frac{1}{8}, \frac{1}{8}, \frac{3}{8})$, $Z(i) = i$, $A_L = \{\{1,2\},\{3,4\}\}$ and $A_R = \{\{1,3\},\{2,4\}\}$. Then

$$(E_{A_R}E_{A_L}Z)(1) = \tfrac{3}{4}\cdot\tfrac{5}{4} + \tfrac{1}{4}\cdot\tfrac{15}{4} = \tfrac{15}{8}, \qquad (E_{A_L}E_{A_R}Z)(1) = \tfrac{3}{4}\cdot\tfrac{3}{2} + \tfrac{1}{4}\cdot\tfrac{7}{2} = 2.$$

$G(A_L, A_R)$ is a 4-cycle, so the meet is trivial. The block $N = \left(\begin{smallmatrix} 3/8 & 1/8 \\ 1/8 & 3/8 \end{smallmatrix}\right)$ has rank two, and K has singular values 1 and $\frac{1}{2}$ (check A3).

3.3 Worked regimes

- *Committed* ($A = B$): the operators coincide. This is a consistency check only.
- *Inherited* ($A \subseteq B$ or $B \subseteq A$): the operators commute with product the coarser one.
- *General*: commutation holds iff (iv) of theorem 3.3 holds. Adding labels does not by itself produce conditional independence. Example 3.5 shows the generic outcome.

3.4 The general case: rate and lax interchange

Theorem 3.6 (Alternating projections, exact rate)**.** *Let c be the cosine of the Friedrichs angle between $L^2(A)$ and $L^2(B)$. In the finite case, c is the largest singular value of K strictly below 1, or 0 if there is none. Then*

$$\|(E_A E_B)^n - E_{A\wedge B}\| = c^{2n-1} \qquad (n \geq 1).$$

On a general probability space the identity holds with the Friedrichs cosine. Then $c < 1$ iff $L^2(A) + L^2(B)$ is closed. Otherwise $(E_A E_B)^n \to E_{A\wedge B}$ only strongly, and the convergence can be arbitrarily slow.

Proof. The identity is the Kayalar–Weinert theorem [21], with $L^2(A) \cap L^2(B) = L^2(A \wedge B)$. Strong convergence is von Neumann's theorem [29], and arbitrarily slow convergence is in [12]. □

Check A4 verifies the rate to 10^{-9} for $n \leq 6$ on 157 non-commuting pairs.

Definition 3.7 (Convex order)**.** For finitely supported laws Q, Q' on a convex set, $Q \leq_{\mathrm{cx}} Q'$ if $\int \varphi\, dQ \leq \int \varphi\, dQ'$ for all convex φ. By Strassen's theorem [28] this holds iff $Q = \mathrm{Law}(\mathbb{E}[U' \mid G])$ for some $U' \sim Q'$ and some sub-σ-algebra G.

Corollary 3.8 (Lax interchange)**.** $E_{A\wedge B}E_A E_B = E_{A\wedge B} = E_{A\wedge B}E_B E_A$. *Hence*

$$\mathrm{Law}(E_{A\wedge B}Z) \leq_{\mathrm{cx}} \mathrm{Law}(E_A E_B Z) \leq_{\mathrm{cx}} \mathrm{Law}(E_B Z) \leq_{\mathrm{cx}} \mathrm{Law}(Z),$$

and symmetrically through $E_B E_A Z$. The laws $\mathrm{Law}((E_A E_B)^n Z)$ decrease in convex order, and

$$W_2\big(\mathrm{Law}((E_A E_B)^n Z), \mathrm{Law}(E_{A\wedge B}Z)\big) \leq c^{2n-1}\|Z - E_{A\wedge B}Z\|_{L^2(p)}.$$

Proof. The tower property gives the first identity. Each displayed law is that of a conditional expectation of the next on the same space. For the rate, $(E_A E_B)^n - E_{A\wedge B} = ((E_A E_B)^n - E_{A\wedge B})(I - E_{A\wedge B})$; apply theorem 3.6 and use the identity coupling to bound W_2. □

So without nestedness or conditional independence, interchange does not hold strictly. It holds through the cospan $E_A E_B \Rightarrow E_{A\wedge B} \Leftarrow E_B E_A$ of convex-order comparisons. These are identities exactly under theorem 3.3, and they shrink at the rate of theorem 3.6.

3.5 Conditioning on a third variable

Proposition 3.9. *Let $X_L \perp X_R \mid H$ on a finite space with full support. Then:*

(a) $E_{\sigma(X_L,H)} E_{\sigma(X_R,H)} = E_{\sigma(H)} = E_{\sigma(X_R,H)} E_{\sigma(X_L,H)}$*;*

(b) $E_{\sigma(X_L)}$ *and* $E_{\sigma(X_R)}$ *commute iff* $X_L \perp X_R$ *marginally. With full support the meet is trivial, so this fails as soon as both channels depend on* H*.*

Proof. (a) is lemma 3.2 with $C = \sigma(H)$. (b) With full support, $G(\sigma(X_L), \sigma(X_R))$ is complete bipartite, hence connected; apply theorem 3.3. □

Conditional independence given a third variable is therefore not a commutation hypothesis. The conditioning variable must be built into both σ-algebras (check A7).

4 Shadows

4.1 Label-defined shadows on laws

Let $S = \mathsf{D}B$ with its Borel σ-algebra, and let $a\colon S \to I_a$ be a measurable *label map* with finite codomain. For $\nu \in \mathsf{D}(S)$ with identity variable Z, put $m_a = \mathbb{E}_\nu[Z \mid \sigma(a)]$ and define the label shadow $P_a(\nu) = (m_a)_*\nu$.

Proposition 4.1 (Properties of a single label shadow)**.** *(a)* $\int \rho\, dP_a(\nu) = \int \rho\, d\nu$*, and* $P_a(\nu) \le_{\mathrm{cx}} \nu$*.*

(b) If $a(m_a(\rho)) = a(\rho)$ *for* ν*-a.e.* ρ *(*label preservation*), then* $P_a(P_a\nu) = P_a\nu$*. Without label preservation, the first barycentre can move a point into another label class, and idempotence can fail.*

Proof. (a) is the tower property together with Strassen's theorem. (b) Write $Y = m_a(Z)$. Then Y is $\sigma(a(Z))$-measurable, and label preservation gives $a(Y) = a(Z)$, so $\mathbb{E}[Y \mid a(Y)] = Y$. □

Proposition 4.2 (Composites of label shadows are single poolings)**.** *For label maps* a, b*,*

$$P_b(P_a(\nu)) = \mathrm{Law}\left(E_{A'} Z\right), \qquad A' = \sigma(b \circ m_a) \subseteq \sigma(a).$$

So $P_b P_a$ *pools once, onto a coarsening of the* first *partition. The equation* $P_b P_a = P_a P_b$ *compares coarsenings of two different partitions. It is not the operator identity* $E_A E_B = E_B E_A$*, and theorem 3.3 does not govern it.*

Proof. m_a is constant on $\sigma(a)$-cells, so $b \circ m_a$ is too. Pooling the atoms of $P_a\nu$ along b merges whole cells, and a barycentre of barycentres is the barycentre of the union. □

Example 4.3 (Threshold labels)**.** On $\mathsf{D}\{0,1\} \cong [0,1]$, let ν be uniform on $\{0.1, 0.3, 0.7, 0.9\}$, $a = [s \ge \frac{1}{2}]$ and $b = [s \ge \frac{1}{4}]$. Then $P_b(P_a\nu) = \frac{1}{2}\delta_{0.2} + \frac{1}{2}\delta_{0.8}$, whereas $P_a(P_b\nu) = \frac{1}{4}\delta_{0.1} + \frac{3}{4}\delta_{19/30}$. The fixed-space operators also fail to commute, but with different values (checks C2, C3).

4.2 Labelled random measures

Definition 4.4. A *labelled random measure* over X is a triple (I, p, Z): a finite probability space and a map $Z\colon I \to \mathsf{D}X$. A morphism $f\colon (I,p,Z) \to (I',p',Z')$ is a map with $f_*p = p'$ and $Z' \circ f = Z$. The functor $\mathrm{Law}(I,p,Z) = Z_*p \in \mathsf{D}^2X$ forgets the labels.

Proposition 4.5 (Shadows on a fixed space)**.** *For partitions A, B of I:*

(a) E_A is idempotent, preserves $\mathbb{E}Z$, and satisfies $\mathrm{Law}(E_AZ) \le_{\mathrm{cx}} \mathrm{Law}(Z)$;

(b) composition of shadows is composition of conditional expectations;

(c) $\mathrm{Law}(E_AE_BZ) = \mathrm{Law}(E_BE_AZ)$ for all Z iff (i)–(v) of theorem 3.3 hold (when $|X| \ge 2$).

Proof. (a) and (b) are immediate. For "only if" in (c): the energy $\int \|v\|^2\, d\,\mathrm{Law}(\cdot)$ is a function of the law. Write $Z = \rho_0 + Y$ with ρ_0 in the relative interior and Y small with zero coordinate sums. Then $\|E_AE_BY\| = \|E_BE_AY\|$ on an open set, hence everywhere, and polarisation gives $S := E_BE_AE_B = E_AE_BE_A$. The range of S lies in $L^2(M)$ with $M = A \wedge B$, and S is self-adjoint, so $S = E_MSE_M = E_M$. For $Y \in L^2(B)$ this reads $\|E_AY\| = \|E_MY\|$, so $E_AY = E_MY$ by Pythagoras, which is (ii). □

Proposition 4.6 (Naturality)**.** *Let $f\colon (I,p) \to (I',p')$ be measure preserving, and let $A = f^{-1}A'$, $B = f^{-1}B'$. Then $E_A(Z' \circ f) = (E_{A'}Z') \circ f$, and likewise for B.*

Proof. $f^{-1}(a')$ is a cell of A of mass $p'(a')$, and the average of $Z' \circ f$ over it is the average of Z' over a' (check G2). □

4.3 The star space of a context and the canonical shadows

Definition 4.7 (Star space)**.** For a coherent context c, let $\kappa(\theta \mid r) = \Sigma(\theta, r)/\Pi(r)$. The *star space* of c has points $(\widetilde{\Pi}, r, \theta, x)$ with weights

$$p_c(\widetilde{\Pi}, r, \theta, x) = \Omega(\widetilde{\Pi})\, \widetilde{\Pi}(r)\, \kappa(\theta \mid r)\, r(x),$$

restricted to positive weights. Write $\tilde{r}$ for the coordinate r and $Z = \delta_x$.

Given $\tilde{r}$, the three *leaves* $\widetilde{\Pi}$, θ and x are independent. The marginals recover the context: $\mathrm{Law}(\widetilde{\Pi}) = \Omega$, $\mathrm{Law}(\tilde{r}) = \Pi$, $\mathrm{Law}(\theta, \tilde{r}) = \Sigma$ and $\mathrm{Law}(x) = \rho$ (check B1). The context itself does not couple Ω with Σ; the conditional product over $\tilde{r}$ is the canonical coupling.

Definition 4.8 (Canonical shadows)**.** For a set $T \subseteq \{\mathrm{lev2}, \mathrm{hyp}, \mathrm{obs}\}$ of leaves, let

$$G_T = \sigma(\tilde{r},\ \text{leaves in } T), \qquad S_T = E_{G_T}.$$

The *left shadow* is $L_c = S_{\{\mathrm{lev2}\}}$ and the *right shadow* is $R_c = S_{\{\mathrm{hyp}\}}$.

Theorem 4.9 (The shadow algebra is a commutative band)**.** *$S_TS_U = S_{T\cap U} = S_US_T$ for all leaf sets T, U. In particular $L_cR_c = R_cL_c = E_{\sigma(\tilde{r})}$.*

Proof. $G_{T\cap U} \subseteq G_T \cap G_U$. The leaves in $T \setminus U$ and in $U \setminus T$ are independent given $\tilde{r}$ and the leaves in $T \cap U$, so lemma 3.2 applies. □

Check B2 verifies all 8×8 products on 60 random coherent contexts. The homomorphism $T \mapsto S_T$ need not be injective: if θ is a function of $\tilde{r}$, then $R_c = S_\emptyset$.

Proposition 4.10 (Dropping the centre)**.** *The label-only shadows $E_{\sigma(\widetilde{\Pi})}$ and $E_{\sigma(\theta)}$ commute iff theorem 3.3 holds for them. They fail to commute in 40 of 60 random contexts (check B3).*

Proposition 4.11 (Markov absorption)**.** *For $Z = \delta_x$: (a) $R_c Z = E_{\sigma(\tilde{r})} Z = \tilde{r}$; (b) $E_{\sigma(\theta)} Z = \sum_r \Sigma(r \mid \theta)\, r$, and in general $E_{\sigma(\tilde{r})} E_{\sigma(\theta)} Z \neq E_{\sigma(\theta)} Z$ (59 of 60 random contexts, check B5).*

Proof. (a) holds because $x \perp \theta \mid \tilde{r}$; (b) is a direct computation. □

The hypothesis level is bookkeeping: given the predictive, it does not move the prediction.

5 Topology and order: how the tower acts

5.1 Two idempotents

The unit laws $\mu \circ \eta_{\mathsf{D}} = \mathrm{id} = \mu \circ \mathsf{D}(\eta)$ give one retraction $\mu \colon \mathsf{D}^2 X \to \mathsf{D}X$ with two sections, and hence two split idempotents on $\mathsf{D}^2 X$:

$$e_\perp = \eta_{\mathsf{D}X} \circ \mu \;\; (\Pi \mapsto \delta_{\mu\Pi}), \qquad e_\top = \mathsf{D}(\eta_X) \circ \mu \;\; (\Pi \mapsto \textstyle\sum_x (\mu\Pi)(x)\, \delta_{\delta_x}).$$

Proposition 5.1 (Two idempotents)**.** *(a) $e_\perp$ and $e_\top$ are idempotent, natural in X, and split through the same retraction μ.*

(b) They form a left-zero band: $e_\perp e_\top = e_\perp$ and $e_\top e_\perp = e_\top$.

(c) $e_\top$ is affine; $e_\perp$ is not.

(d) For W_1 built from ℓ^1: μ is 1-Lipschitz, η and $\mathsf{D}(\eta)$ are isometries, and $e_\perp$, $e_\top$ are 1-Lipschitz, hence weakly continuous.

Proof. (a) These are the unit laws and naturality of η and μ. (b) $\eta\mu\mathsf{D}(\eta)\mu = \eta\mu$ and $\mathsf{D}(\eta)\mu\eta\mu = \mathsf{D}(\eta)\mu$. (c) μ and $\mathsf{D}(\eta)$ are affine. η is not, since $\delta_{s\rho+(1-s)\rho'} \neq s\delta_\rho + (1-s)\delta_{\rho'}$. (d) $r \mapsto \langle g, r\rangle$ is 1-Lipschitz for $g \in [-1,1]^X$, so $\|\mu\Pi - \mu\Pi'\|_1 \leq W_1(\Pi, \Pi')$. Also $W_1(\delta_\rho, \delta_{\rho'}) = \|\rho - \rho'\|_1$, and distinct vertices are at distance 2, so $W_1(\mathsf{D}(\eta)\rho, \mathsf{D}(\eta)\rho') = \|\rho - \rho'\|_1$. □

The pooling idempotent of the tower paper is $e = e_\perp$ [1, §13.6], [3].

5.2 Barycentre fibres

Proposition 5.2 (Fibres are convex-order intervals)**.** *For $\rho \in \mathsf{D}X$, let $F_\rho = \mu^{-1}(\rho)$ be the set of Borel probability measures on Δ_X with barycentre ρ. Then:*

(a) F_ρ is convex and weakly compact, and its finitely supported elements are dense in it;

(b) $e_\perp \Pi = \delta_\rho \leq_{\mathrm{cx}} \Pi \leq_{\mathrm{cx}} \mathsf{D}(\eta)\rho = e_\top \Pi$ for every $\Pi \in F_\rho$;

(c) $\mathsf{D}(\eta)\rho$ is the unique element of F_ρ supported on the vertices.

Proof. (a) Prokhorov's theorem and continuity of μ give compactness; F_ρ is convex because μ is affine. For density, partition Δ_X into cells of diameter $< \varepsilon$ and pool Π within each cell. This keeps the barycentre and moves Π by less than ε in W_1. (b) $\rho \mapsto r \sim \Pi$ and $r \mapsto \delta_x$ with probability $r(x)$ are martingale couplings (check B6). (c) Vertices are affinely independent. This is the Choquet–Meyer uniqueness of maximal measures on a simplex [27]. □

5.3 A coherent context is a martingale

Theorem 5.3 (The tower as a martingale)**.** *On the star space let $\mathcal{F}_0$ be trivial, $\mathcal{F}_1 = \sigma(\widetilde{\Pi})$, $\mathcal{F}_2 = \sigma(\widetilde{\Pi}, \tilde{r})$ and $\mathcal{F}_3 = \sigma(\widetilde{\Pi}, \tilde{r}, x)$, and put $M_k = \mathbb{E}[Z \mid \mathcal{F}_k]$. Then (M_k) is a Δ_X-valued martingale with*

$$\mathrm{Law}(M_0) = \delta_\rho, \quad \mathrm{Law}(M_1) = \mathsf{D}(\mu)\Omega, \quad \mathrm{Law}(M_2) = \Pi, \quad \mathrm{Law}(M_3) = \mathsf{D}(\eta)\rho,$$

a chain $\delta_\rho \leq_{\mathrm{cx}} \mathsf{D}(\mu)\Omega \leq_{\mathrm{cx}} \Pi \leq_{\mathrm{cx}} \mathsf{D}(\eta)\rho$ in F_ρ. The hypothesis branch does not move it: $\mathbb{E}[Z \mid \theta, \tilde{r}] = M_2$.

Proof. $M_0 = \rho$, $M_1 = \mu(\widetilde{\Pi})$, $M_2 = \tilde{r}$ and $M_3 = \delta_x$. The last statement is proposition 4.11 (check B4). □

So the levels of the tower are the successive conditional expectations of the observation along one filtration. Coherence is the martingale property in law. Nested shadows land on levels and commute.

5.4 The Beck defects located

Proposition 5.4 (Where the defects of λ^σ sit)**.** *At level one:*

(a) the two sides of (U) at (x, c) are $e_\perp \Pi = \delta_\rho$ and $e_\top \Pi = \mathsf{D}(\eta)\rho$, so the unit defect is the full convex-order diameter of F_ρ;

(b) for $M' = \sum_\ell \nu_\ell \delta_{M_\ell}$ with $M_\ell = \sum_k a_{\ell k} \delta_{(x_{\ell k}, c_{\ell k})}$, on the sample-index space $I = \{(\ell, k)\}$ with $p = \nu_\ell a_{\ell k}$ and $W(\ell, k) = \rho_{\ell k}$, the two sides of (M) are $\mathrm{Law}(W)$ (λ-first) and $\mathrm{Law}(\mathbb{E}[W \mid \ell])$.

Proof. This is [1, Prop. 13.8] read in the present notation, and it is verified with the tower code's own λ^σ, C_3, η and μ (check B7). □

The shift law pools after the unit and spreads after the multiplication. Every order that reads conditional expectation monotonically therefore orients the two defects oppositely [1, Prop. 13.9, Thm. 13.15, Prop. 13.17].

5.5 Continuity

Proposition 5.5. *(a) On a fixed index with fixed partitions, $(p, Z) \mapsto \mathrm{Law}(E_A Z)$ is continuous for full-support p, and $W_1(\mathrm{Law}(E_A Z), \mathrm{Law}(E_A Z')) \leq \|Z - Z'\|_{L^1(p)}$. (b) Nearest-atom shadows are Borel but not continuous. With atoms $(\frac{2}{3}, \frac{1}{3})$ and $(\frac{1}{3}, \frac{2}{3})$, two inputs at W_1-distance $\frac{1}{1000}$ have outputs at distance $\frac{501}{2000}$ (check C4).*

Proof. (a) Cell averages are continuous where the cell masses are positive, and conditional expectation is an L^1 contraction. (b) A point crossing the Voronoi wall changes cells. □

6 Categorical structure

6.1 Why a separate construction is needed

The operator results act on measures and random variables; context maps act on contexts. A double category needs objects, horizontal and vertical arrows with associative compositions, squares with typed boundaries, identity squares, two compositions of squares, and the interchange law for composable grids. Expressions such as $g \circ L$ or $R \circ f$ are undefined unless a functor connects the two layers. This section builds that functor and then the double category.

6.2 The context category

Definition 6.1 (Context category)**. Ctx** has as objects the coherent contexts c on finite bases B_c. A morphism $f\colon c \to c'$ is a map $f\colon B_c \to B_{c'}$ with

$$c' = \big(f_*\rho,\ \mathsf{D}^2 f(\Pi),\ \mathsf{D}^3 f(\Omega),\ \mathsf{Sh} f(\Sigma)\big).$$

Composition and identities are those of maps of finite sets.

Proposition 6.2. Ctx *is a category, and pushing forward along f preserves coherence.*

Proof. D^k and Sh are functors. Coherence is preserved by naturality of μ ($\mu \circ \mathsf{D}^2 f = \mathsf{D} f \circ \mu$) and of q. □

6.3 The lifting functor, in two versions

Remark 6.3 (Two readings of the level-two component)**.** A morphism $f\colon c \to c'$ can carry either of two kinds of level-two data:

(i) a specified map $f_\Pi\colon \mathsf{D}^2(B_c) \to \mathsf{D}^2(B_{c'})$, with $(g \circ f)_\Pi = g_\Pi \circ f_\Pi$ imposed as an axiom (*component-data version*);

(ii) a base map $f\colon B_c \to B_{c'}$, with induced action $\mathsf{D}^2(f)$ (*base-map version*, as in definition 6.1).

The two agree when $f_\Pi = \mathsf{D}^2(f)$. This is a coherence condition, not automatic.

Definition 6.4 (Lifting functor)**.** Define $F(c) = \mathsf{D}^2(B_c)$, with $F(f) = f_\Pi$ in the component-data version and $F(f) = \mathsf{D}^2(f)$ in the base-map version.

Proposition 6.5 (Functoriality, component-data version)**.** *If* $(\mathrm{id}_c)_\Pi = \mathrm{id}$ *and* $(g \circ f)_\Pi = g_\Pi \circ f_\Pi$*, then F is a functor.*

Proof. $F(\mathrm{id}_c) = (\mathrm{id}_c)_\Pi = \mathrm{id}_{F(c)}$ and $F(g \circ f) = (g \circ f)_\Pi = g_\Pi \circ f_\Pi = F(g) \circ F(f)$, and all three maps have type $\mathsf{D}^2(B_c) \to \mathsf{D}^2(B_{c''})$. □

Proposition 6.6 (Functoriality, base-map version)**.** *In the base-map version F is a functor.*

Proof. D^2 is a functor. □

Remark 6.7. In the component-data version f_Π is already a map between level-two spaces. Applying D^2 to it again would give a map $\mathsf{D}^4(B_c) \to \mathsf{D}^4(B_{c'})$ of the wrong type, so $F(f) = f_\Pi$ is the type-correct choice there.

6.4 Shadows along context maps

The canonical shadows live on star spaces, not on $F(c)$. The star construction is functorial on a large subcategory.

Proposition 6.8 (Star spaces are functorial)**.** *Let $f\colon c \to c'$ be a morphism of* **Ctx** *such that* $\mathsf{D} f$ *is injective on* $\operatorname{supp}\Pi$*. Then*

$$\tilde{f}(\widetilde{\Pi}, r, \theta, x) = \big(\mathsf{D}^2 f(\widetilde{\Pi}),\ \mathsf{D} f(r),\ \theta,\ f(x)\big)$$

maps the star space of c measure-preservingly onto the star space of c'. It pulls back G_T to G_T for $T \subseteq \{\mathrm{lev2}, \mathrm{hyp}\}$. Hence L_c and R_c are natural: $L_c(Z' \circ \tilde{f}) = (L_{c'} Z') \circ \tilde{f}$, and likewise for R.

Proof. Summing p_c over the fibre of $\tilde{f}$ gives $\Omega(\widetilde{\Pi})\widetilde{\Pi}(r)\kappa(\theta \mid r)\sum_{x\in f^{-1}(x')} r(x)$. By injectivity on atoms, each factor equals its primed counterpart. Injectivity also gives $\tilde{f}^{-1}\sigma(\mathsf{D}f(\tilde{r})) = \sigma(\tilde{r})$ and $\tilde{f}^{-1}\sigma(\mathsf{D}^2 f(\widetilde{\Pi})) = \sigma(\widetilde{\Pi})$, while θ is unchanged. Naturality is proposition 4.6 (check G3, on 36 context maps). For $T \ni \mathrm{obs}$ only $\tilde{f}^{-1}G'_T \subseteq G_T$ holds, because f may merge outcomes. □

So the "(L, R)-compatible subcategory" of a context-map formalism is not hypothetical. It contains every context map that is injective on predictive atoms: relabellings of latent classes, coarsenings of the observation space that keep the atoms distinct, and their composites.

6.5 Typed squares and quintets

Definition 6.9 (Typed square data)**.** A square with boundary ($h\colon a \to b$, $v\colon a \to c$, $w\colon b \to d$, $k\colon c \to d$) may carry a comparison α only between the composites $k \circ v$ and $w \circ h$, which share source and target. The boundary, the comparison, and the composition laws are all part of the data.

Proposition 6.10 (Quintets of a locally preordered 2-category)**.** *Let $\mathcal{K}$ be a 2-category whose hom-categories are preorders. The quintet double category $Q(\mathcal{K})$ has the objects and 1-cells of $\mathcal{K}$ in both directions, and exactly one square with boundary (h, v, w, k) iff $k \circ v \le w \circ h$. Then $Q(\mathcal{K})$ is a strict double category, and interchange, associativity and the unit laws hold automatically.*

Proof. This is Ehresmann's quintet construction [17, 19]. Pasting is transitivity together with monotone whiskering, and uniqueness of 2-cells makes every equation trivial. □

In a locally discrete 2-category such as **Meas**, where the only 2-cells are identities [1, §§13.1, 13.5], squares are commuting squares. Nontrivial cells require an order.

6.6 What the tower supplies, and what it lacks

(a) *Convex order on one level.* $\le_{\mathrm{cx}}$ is a partial order on $\mathsf{D}(K)$ for convex K, and pushforward along affine maps is monotone. $T \mapsto \mathrm{Law}(S_T Z)$ is a monotone map from the Boolean lattice of leaf sets into $(F_\rho, \le_{\mathrm{cx}})$. The martingale of theorem 5.3 is a chain of vertical arrows.

(b) *Strict interchange for canonical shadows* (theorem 4.9).

(c) *Lax interchange in general*: a cospan of squares into the meet (corollary 3.8).

(d) *The Beck defects.* The (U) and (M) defects of λ^σ are vertical arrows of opposite orientation, so λ^σ is neither lax nor oplax in any $Q(\mathcal{K})$ whose order reads conditional expectation monotonically. The (Cm) defect of $\bar{\lambda}$ is incomparable in the joint convex order [1, Prop. 13.10]. It is a cell of one orientation for label pooling at every level [1, Thm. 13.16].

(e) *What is missing.* D must act monotonically on hom-preorders. The pointwise convex order fails this, because $\delta_a \le_{\mathrm{cx}} \delta_b$ forces $a = b$. The candidate is the relation (Barr) lifting of the preorder along D. Since D preserves weak pullbacks, this lifting preserves reflexivity and transitivity and makes D monotone.

Open problem 6.11 (A 2-category for the tower)**.** Are C_3, its structure maps, and the label-pooling cells of $\bar{\lambda}$ monotone for the Barr lifting of label pooling, so that $\bar{\lambda}$ becomes a lax mixed distributive law? This is open problem 4 of [1], now with a definite candidate.

7 Beck defects as comparison cells

7.1 What a factorisation requires

Proposition 7.1 (Required data)**.** *To factor a Beck defect through a shadow, one needs:*

(i) an action of the shadow on the relevant tower objects;

(ii) a comparison map through that action;

(iii) a typed coherence map completing the factorisation;

(iv) a proof that the composite comparison equals the original defect.

The observation that one side looks pooled and the other unpooled motivates a factorisation but does not prove one. For the defects computed so far, all four items are now supplied.

Proposition 7.2 (Each computed defect is one conditional-expectation step)**.** *On a fixed labelled space built from the data of the axiom:*

(a) (U) for λ^σ at level one: the space is the star space, $Z = \delta_x$, and the step is E_{triv}. The sides are $\mathrm{Law}(E_{\mathrm{triv}}Z) = \delta_\rho$ and $\mathrm{Law}(Z) = \mathsf{D}(\eta)\rho$.

(b) (M) for λ^σ at level one: the space is the sample index, and the step is $E_{\sigma(\ell)}$ (proposition 5.4(b)).

(c) (Cm) for $\bar{\lambda}$ at every level and in the shadow: the space is the sample index, and the step is the coordinatewise conditional expectation of the context labels onto the trivial σ-algebra, with the sample coordinate kept [1, Thm. 13.16].

In each case items (i)–(iv) of proposition 7.1 hold: the shadow acts on the labelled space, the comparison is the martingale coupling, the coherence map is the identity on the index, and the equality is the cited computation.

Remark 7.3 (The two-shadow conjecture)**.** An earlier formulation conjectured that the (Cm) defect factors through a *pair* of shadows. In every computed case one conditional-expectation step suffices. Whether some Beck defect of a tower variant needs two non-commuting shadows, and so carries a genuinely lax interchange (corollary 3.8), is open.

7.2 One-shadow and two-shadow options

Option A keeps one shadow component, $\mathsf{C}_A(Y) = Y \times \mathsf{D}B \times \mathsf{D}^2B \times \mathsf{D}^3B \times \mathsf{Sh}(Y)$. Option B keeps two, $\mathsf{C}_B(Y) = \mathsf{C}_A(Y) \times \mathsf{Sh}(\mathsf{D}Y)$.

Proposition 7.4. *Let $\mathsf{C}_E Y = Y \times \prod_i E_i Y$ be built from functors E_i, and let λ be assembled from transports $\lambda_i \colon \mathsf{D}E_i \Rightarrow E_i\mathsf{D}$. Then (U), (Co) and (M) hold for λ iff they hold componentwise. (Cm) does not decompose this way, because $\delta(y, c) = ((y, c), \prod_i E_i(j_c)(c_i))$ feeds the* whole *context into each component.*

Proof. The structure maps of D act factorwise, and equality in a product is componentwise. The formula for δ is [1, §6]. □

So neither option repairs a failed axiom automatically. For (Cm) the averaging-type transport has its defect in every component at once [1, Thm. 13.16].

7.3 Higher cells

Proposition 7.5 (Triviality after equality)**.** *If two parallel morphisms are equal, the identity comparison on them is trivial. In a quintet double category of a locally preordered 2-category, any two parallel 2-cells are equal, so no nontrivial 3-cells arise.*

A nontrivial 3-cell would require hom-categories that are not preorders. It would then be a comparison between two distinct comparisons with the same boundary. No such structure is used or needed here.

8 The measure-theoretic (Giry) extension

Let X be Polish and let $\mathcal{P}(X)$ carry the weak topology. Then $\mathcal{P}$ is a monad on Polish spaces, with $\eta_X(x) = \delta_x$ and $\mu_X(\Pi)(A) = \int \rho(A)\, d\Pi(\rho)$, both continuous [18]. On finite discrete X it is D. A category of closed convex subsets of Euclidean spaces is not closed under $\mathcal{P}$, so the continuous tower must use spaces of measures as objects. A measure-valued shadow needs five ingredients, and each is now available:

(i) *a measurable space of measures*: $\mathcal{P}(X)$ with its Borel structure, which is standard Borel when X is;

(ii) *a conditional barycentre*: for $Z\colon \Omega \to \mathcal{P}(X)$ and $G \subseteq \mathcal{F}$, set $\mathbb{E}[Z \mid G](A) = \mathbb{E}[Z(A) \mid G]$; when X is standard Borel this has a version that is a random probability measure;

(iii) *measurability*: the version can be chosen measurable;

(iv) *integrability*: automatic, since values are probability measures;

(v) *structure*: propositions 4.5 and 4.6 hold verbatim on a fixed probability space.

The commutation results extend as follows.

- (i)⇔(ii)⇔(iii) of theorem 3.3 hold for sub-σ-algebras of any probability space, applied to $Z(A)$ for A in a countable generating π-system.
- The finite criteria (iv) and (v) are replaced by the spectral condition $E_A E_B E_A = E_{A\wedge B}$ on $L^2(A)$. The meet need not be countably generated.
- Theorem 3.6 holds with c the Friedrichs cosine. If $c = 1$, the lax interchange holds only in the strong limit.
- For compact metric X, proposition 5.2(a,b) and theorem 5.3 hold with $\mathcal{P}(\eta)\rho$ as top. Since $\mathcal{P}(X)$ is a Bauer simplex, maximal representing measures are unique [27].

Open problem 8.1 (Lax interchange without a rate)**.** When $c = 1$, does the cospan of corollary 3.8 admit a coherent choice, natural along measure-preserving maps that do not pull back the σ-algebras?

9 The finite stochastic-to-convex functor

Definition 9.1. FinStoch has as objects finite sets and as morphisms stochastic kernels $K\colon X \to Y$, i.e. $K(y \mid x) \geq 0$ with $\sum_y K(y \mid x) = 1$. Composition is $(L \circ K)(z \mid x) = \sum_y L(z \mid y)K(y \mid x)$, and the identities are Kronecker deltas. It is the Kleisli category of D restricted to finite sets. **Conv** has as objects the simplices Δ_X and as morphisms the affine maps. Every kernel induces

$$T_K\colon \Delta_X \to \Delta_Y, \qquad T_K(p)(y) = \sum_x p(x)K(y \mid x). \tag{2}$$

Lemma 9.2 (Well-definedness and affineness)**.** *$T_K(p) \in \Delta_Y$, and $T_K(tp + (1-t)q) = tT_K(p) + (1-t)T_K(q)$.*

Proof. Every term is nonnegative, and $\sum_y T_K(p)(y) = \sum_x p(x) \sum_y K(y \mid x) = 1$. Affineness holds because (2) is linear in p. □

Lemma 9.3 (Identities and composition)**.** *$T_{I_X} = \mathrm{id}$ and $T_{L\circ K} = T_L \circ T_K$.*

Proof. $T_{I_X}(p)(x') = \sum_x p(x)\delta_{xx'} = p(x')$. Also $T_{L\circ K}(p)(z) = \sum_x p(x) \sum_y L(z \mid y)K(y \mid x) = \sum_y T_K(p)(y)L(z \mid y)$. □

Theorem 9.4 (Fully faithful functor)**.** *$T\colon$ **FinStoch** $\to$ **Conv** is a fully faithful functor. Each T_K is continuous and ℓ^1-non-expansive: $\|T_K(p) - T_K(q)\|_1 \leq \|p - q\|_1$.*

Proof. Functoriality is lemmas 9.2 and 9.3. An affine map of simplices is determined by the images of the vertices, which may be any points, so the affine maps $\Delta_X \to \Delta_Y$ are exactly the stochastic matrices. This is the comparison functor from the Kleisli to the Eilenberg–Moore category, restricted to free algebras. Non-expansiveness: $\sum_y |\sum_x (p-q)(x)K(y \mid x)| \leq \sum_x |(p-q)(x)|$ (checks E1, E2). □

Relation to the tower. T sends B to Δ_B and a kernel to an affine map, and a Beck diagram of kernels commutes in **FinStoch** iff it commutes in **Conv**, because T is faithful. Two cautions apply.

- The iterated levels $\mathsf{D}^2B, \mathsf{D}^3B$ are infinite-dimensional convex sets, not finite simplices.
- The shadows are not affine (proposition 5.1(c)), so they are not in the image of T, and their interchange is not a commutation of affine maps.

10 A Gaussian model: fusion and its type-correct domain

Let $B = \{A, B, C\}$ with uniform prior π. Let X_L, X_R be real channels, conditionally independent given the class $H = k$:

$$X_L \mid H = k \sim N(\mu_{L,k}, 1), \qquad X_R \mid H = k \sim N(\mu_{R,k}, 1),$$

with $\mu_{L,A} = -2$, $\mu_{L,B} = 0$, $\mu_{L,C} = 2$, $\mu_{R,A} = \mu_{R,C} = -1$ and $\mu_{R,B} = 1$. The left channel separates A from C; the right channel cannot. The level-one posteriors $Q_L(x)$, $Q_R(y)$ and $Q_{LR}(x,y)$ take values in Δ_B.

Proposition 10.1 (Fusion)**.** *$Q_{LR}(x,y) = \mathrm{Fuse}(Q_L(x), Q_R(y))$, where $\mathrm{Fuse}(q,r)_k = \dfrac{q_k r_k/\pi_k}{\sum_j q_j r_j/\pi_j}$.*

Proof. $Q_{LR} \propto \pi_k \ell_L(x \mid k)\ell_R(y \mid k)$, and $Q_L Q_R/\pi \propto \pi_k \ell_L(x \mid k)\ell_R(y \mid k)$. This holds exactly under conditional independence (check G1, discrete channels, exact). □

Domain. Fuse is defined on $U = \{(q, r) : \sum_j q_j r_j / \pi_j > 0\}$. Gaussian likelihoods are strictly positive, so $Q_L(x)$ and $Q_R(y)$ lie in the open simplex, and (Q_L, Q_R) maps all of $\mathbb{R}^2$ into U. The fusion square

$$\begin{array}{ccc} \mathbb{R}^2 & \xrightarrow{(Q_L,Q_R)} & U \\ & \searrow^{Q_{LR}} & \downarrow \mathrm{Fuse} \\ & & \Delta_B \end{array}$$

is therefore total and commutes everywhere. No restriction of the domain is needed. The pushforward $(Q_L, Q_R)_* P_{XY}$ has support touching the boundary of $\Delta_B \times \Delta_B$ only in closure, and the boundary has measure zero. The level-two laws are $S_L = (Q_L)_* P_X$, $S_R = (Q_R)_* P_Y$ and $S_{LR} = (Q_{LR})_* P_{XY}$, and the pushforward of the square along D commutes.

Shadows in the Gaussian model. By proposition 3.9, the shadows that commute are $E_{\sigma(X_L,H)}$ and $E_{\sigma(X_R,H)}$, with product $E_{\sigma(H)}$. The shadows $E_{\sigma(X_L)}$ and $E_{\sigma(X_R)}$ do not commute: their meet is trivial, because the joint density is positive, and X_L, X_R are dependent through H. A relabelling of the latent classes is a measure-preserving bijection that pulls back all three σ-algebras, so both commuting shadows are natural along it (proposition 4.6).

What the model shows. It gives a finite-dimensional, measurable pair of commuting shadows, a total fusion square, and a non-identity natural symmetry. It does not construct the Beck comparison maps. Nor does it show interchange for shadows that omit H; that interchange is false.

11 Positive tropicalisation and the Berkovich factorisation

11.1 Why a real non-Archimedean field

A non-Archimedean absolute value cannot restrict to the usual one on $\mathbb{R}$, so real probability vectors have no informative Berkovich points. For geometry to appear, probabilities must carry *scale*: families $p(t)$ whose coordinates behave like $c\,t^a$ as $t \to 0$.

Definition 11.1. $K = \mathbb{R}((t^{\mathbb{R}}))$ is the field of real Hahn series $\sum_{a \in S} c_a t^a$ with S well ordered. It is real closed and complete for $\mathrm{val}(f) = \min S$, with value group $\mathbb{R}$. Write $|f| = e^{-\mathrm{val}\, f}$. f is *positive* if its leading coefficient is positive. The real Puiseux series form a subfield, on which all exact checks are carried out.

The one fact behind everything that follows is this:

$$f, g > 0 \implies \mathrm{val}(f + g) = \min(\mathrm{val}\, f, \mathrm{val}\, g), \qquad \mathrm{val}(fg) = \mathrm{val}\, f + \mathrm{val}\, g. \tag{3}$$

So $\mathrm{val}\colon (K_{>0}, +, \times) \to (\mathbb{R}, \min, +)$ is a semiring homomorphism. This is Maslov dequantisation, the Laplace principle at finite level. A positive K-point is a family of distributions together with its large-deviation rates, and val keeps the rates.

11.2 The tropical tower

Definition 11.2 (Two monads)**.** $\mathsf{D}_K X$ is the set of finitely supported $p\colon X \to K_{\geq 0}$ with $\sum p = 1$. $\mathsf{D}_{\mathbb{T}} X$ is the set of finitely supported $v\colon X \to \mathbb{R} \cup \{\infty\}$ with $\min v = 0$, with unit $\eta_{\mathbb{T}}(x) = (0$ at x, ∞ elsewhere) and multiplication $\mu_{\mathbb{T}}(V)(x) = \min_u (V(u) + u(x))$.

Theorem 11.3 (Valuation is a monad morphism)**.** $\mathsf{D}_{\mathbb{T}}$ *is a monad, and* $\mathrm{val}\colon \mathsf{D}_K \Rightarrow \mathsf{D}_{\mathbb{T}}$ *is a morphism of monads:*

$$\mathrm{val} \circ \eta = \eta_{\mathbb{T}}, \qquad \mathrm{val}_X \circ \mu_X = \mu_{\mathbb{T},X} \circ \mathrm{val}_{\mathsf{D}_{\mathbb{T}}X} \circ \mathsf{D}_K(\mathrm{val}_X).$$

Proof. $\mathsf{D}_{\mathbb{T}}$ is the normalised distribution monad of the semiring $(\mathbb{R} \cup \infty, \min, +)$, and $\min_x \min_u (V(u) + u(x)) = \min V = 0$. The map val lands in $\mathsf{D}_{\mathbb{T}}$ because $\sum p = 1$ forces $\min \mathrm{val}\, p = 0$ by (3). The multiplication square is (3) applied to $\sum_\rho \Pi(\rho)\rho(x)$; atoms merged by $\mathsf{D}_K(\mathrm{val})$ have summed weights, whose valuation is the minimum (checks T1, T2). □

Corollary 11.4 (The tropical tower)**.** val *maps coherent* K*-contexts to coherent tropical contexts,* $e_\perp$ *and* $e_\top$ *to their tropical counterparts, and pooling on a cell to the tropical cell barycentre* $\min_j(\mathrm{val}\, p_j + \mathrm{val}\, Z_j) - \min_j \mathrm{val}\, p_j$.

11.3 Fibres

Definition 11.5. For positive $\rho \in \mathsf{D}_K X$, the *k-fibre* $F^{(k)}_\rho$ is the set of positive decompositions $\rho = \sum_{i \le k} w_i r_i$. For $v \in \mathsf{D}_{\mathbb{T}} X$, the *tropical k-fibre* $F^{(k)}_{\mathbb{T},v}$ is the set of $(w, u_1, \dots, u_k)$ with $w \in \mathsf{D}_{\mathbb{T}}[k]$ finite, $u_i \in \mathsf{D}_{\mathbb{T}} X$, and $\min_i(w_i + u_i(x)) = v(x)$ for all x.

Theorem 11.6 (Fibre lifting)**.** $\mathrm{val}(F^{(k)}_\rho) = F^{(k)}_{\mathbb{T},\mathrm{val}\,\rho}$.

Proof. $\subseteq$ is theorem 11.3. For $\supseteq$, let $\alpha(x) = \{i : w_i + u_i(x) = v(x)\}$ be the attainers at x. For $i \notin \alpha(x)$, choose positive $q_i(x)$ with $\mathrm{val}\, q_i(x) = w_i + u_i(x) > v(x)$. The remainder $m(x) = \rho(x) - \sum_{i \notin \alpha(x)} q_i(x)$ has the leading term of $\rho(x)$; split it equally among the attainers. Then $\sum_i q_i(x) = \rho(x)$ exactly. Put $w_i^K = \sum_x q_i(x)$ and $r_i = q_i / w_i^K$. Then $\mathrm{val}\, w_i^K = w_i + \min u_i = w_i$ and $\mathrm{val}\, r_i = u_i$. Check T3 runs this construction on 300 random points, with 83 tied coordinates. □

Proposition 11.7. $F^{(k)}_{\mathbb{T},v}$ *is a finite union of polyhedra, indexed by assignment types* $\alpha\colon X \to 2^{[k]} \setminus \{\emptyset\}$*. It contains the images of the two ends of the classical fibre:* $e_\perp \mapsto (0; v)$ *and* $e_\top \mapsto (w_x = v(x); u_x = \eta_{\mathbb{T}}(x))$.

11.4 Tropical secants

Definition 11.8. A model $V = \varphi(K^d_{>0}) \subseteq \mathsf{D}_K X$ is *positively parametrised* if the coordinates of φ are ratios of polynomials with nonnegative integer coefficients. Its positive tropicalisation is $\mathrm{trop}^+ V = \varphi_{\mathbb{T}}(\mathbb{R}^d)$. The k-mixture is $\mathrm{Mix}_k(V) = \{\sum_{i \le k} w_i r_i\}$, the positive part of the k-th secant variety, and its tropical analogue is $\mathrm{Mix}^{\mathbb{T}}_k(L) = \{\min_i(w_i + u_i)\}$.

Theorem 11.9 (Mixtures tropicalise to tropical secants)**.** $\mathrm{val}(V) = \mathrm{trop}^+ V$ *and* $\mathrm{val}(\mathrm{Mix}_k V) = \mathrm{Mix}^{\mathbb{T}}_k(\mathrm{trop}^+ V)$*, exactly. For a toric model* V_A*,* $\mathrm{trop}^+ V_A$ *is the row space of* A*, normalised, and* $\mathrm{Mix}^{\mathbb{T}}_k$ *is Develin's tropical secant of a linear space [14].*

Proof. $\mathrm{val}\, \varphi(\theta) = \varphi_{\mathbb{T}}(\mathrm{val}\, \theta)$ by (3), and val maps $K^d_{>0}$ onto $\mathbb{R}^d$. For mixtures, apply theorem 11.3 and choose lifts with prescribed valuations. □

Over an algebraically closed field, cancellation can make $\mathrm{trop}(\mathrm{Sec}_k)$ larger than the tropical secant [14, 15]. On the positive part there is no discrepancy. For matrices, the positive tropicalisation of the nonnegative-rank-$\le k$ locus is the Barvinok rank-$\le k$ locus. For real semialgebraic sets the image of val is Alessandrini's logarithmic limit set [11].

Example 11.10 (Dimensions)**.** Generic Jacobian ranks (floating point, tolerance 10^{-6}, 40 points; checks T6, T7):

model	k	$\dim \mathrm{Mix}_k^{\mathbb{T}}$	$\dim \mathrm{Mix}_k$ (affine cone)
3×3 independence	1, 2, 3	5, 8, 9	5, 8, 9
$2 \times 2 \times 2$ independence	1, 2	4, 8	4, 8

In each case the tropical dimension certifies the classical one, in the spirit of [16].

11.5 The Berkovich square

$(\mathbb{A}^m_K)^{\mathrm{an}}$ is the space of multiplicative seminorms on $K[z]$ extending $|\cdot|$ [13]. A K-point y gives the rigid point $|f|_y = |f(y)|$. On the torus there are two maps:

- the skeleton section $s(v)$, the Gauss point $|\sum_m a_m z^m|_{s(v)} = \max_m |a_m| e^{-\langle m, v\rangle}$;
- the retraction $\tau(\xi) = (-\log |z_j|_\xi)_j$.

They satisfy $\tau s = \mathrm{id}$, and $\tau(y) = \operatorname{val} y$ on rigid points. Payne's theorem [26] identifies the analytification with the inverse limit of tropicalisations, and Jell, Scheiderer and Yu prove the real analogue for semialgebraic sets [20].

Theorem 11.11 (Berkovich square)**.** *Let $\Phi\colon \mathbb{A}^m \to \mathbb{A}^n$ be polynomial with integer coefficients, for instance the k-mixture map, and assume the residue characteristic is 0. Then:*

(a) $\tau \circ \Phi^{\mathrm{an}} \circ s = \Phi_{\mathbb{T}}$ on the skeleton;

(b) $\tau \circ \Phi^{\mathrm{an}} = \Phi_{\mathbb{T}} \circ \tau$ on positive rigid points, if Φ has nonnegative coefficients;

(c) $\tau \circ \Phi^{\mathrm{an}} \geq \Phi_{\mathbb{T}} \circ \tau$ on all rigid points, strictly exactly where leading terms cancel.

$$\begin{array}{ccc} F_\rho^{(k)} \subseteq (\mathit{params})_K^{>0} & \xrightarrow{\ \Phi\ } & \mathrm{Mix}_k(V) \subseteq \mathsf{D}_K X \\ \Big\downarrow{\scriptstyle \mathit{rigid}} & & \Big\downarrow{\scriptstyle \mathit{rigid}} \\ (\mathbb{A}^m)^{\mathrm{an}} & \xrightarrow{\ \Phi^{\mathrm{an}}\ } & (\mathbb{A}^n)^{\mathrm{an}} \\ {\scriptstyle s}\big(\ {\scriptstyle \tau}\big\downarrow & & \Big\downarrow{\scriptstyle \tau} \\ F_{\mathbb{T},v}^{(k)} \subseteq \mathbb{R}^m & \xrightarrow[\ \Phi_{\mathbb{T}}\ ]{} & \mathrm{Mix}_k^{\mathbb{T}} \subseteq \mathsf{D}_{\mathbb{T}} X \end{array}$$

Proof. (a) $-\log |\Phi_j|_{s(v)} = \min_m \langle m, v\rangle$, because nonzero integers have absolute value 1 (check T4). (b) is (3). (c) is the ultrametric inequality. □

11.6 What the analogies do not give

A Berkovich point is a multiplicative seminorm, not a probability measure. A rational domain is an algebraic localisation, not conditioning. The Kleisli category is not a site without specified covers. A distributive law is not a structure sheaf. Shadows are conditional expectations, not Grothendieck topologies. Two constructions that tensor $C(\Delta_X)$ with a non-Archimedean field fail for definite reasons:

- $\mathbb{R}$ has no isometric embedding into such a field;

- $C(\Delta_X)$ is not affinoid;
- over $\mathbb{R}$, Gel'fand duality returns Δ_X and the functor T of theorem 9.4, with nothing new.

What *does* hold is theorem 11.11. Positivity, through (3), is the hypothesis that turns the analogy into a theorem.

12 Shadows at zero temperature and secant geometry

12.1 The posterior tropicalises to the hard assignment

Proposition 12.1. *For positive $\rho = \sum_i w_i r_i$, the posterior $P(i \mid x) = w_i r_i(x)/\rho(x)$ has* $\operatorname{val} P(i \mid x) = w_i^{\mathbb{T}} + u_i(x) - v(x) \geq 0$, *with equality iff i attains the minimum. For convergent series, its $t \to 0$ limit is* $\operatorname{lc}(w_i r_i(x))/\operatorname{lc}(\rho(x))$ *on the attainers and* 0 *elsewhere, where* lc *is the leading coefficient (check T5).*

So conditioning on the observation tropicalises to argmin (MAP) assignment. Nearest-atom pooling is a *zero-temperature shadow*, and its Voronoi walls are tie loci. The same mechanism gives k-means as the small-variance limit of Gaussian mixtures [22].

Proposition 12.2 (Walls)**.** *(a) $\mu_{\mathbb{T}}$ is affine on each assignment cell. Crossing a tie hyperplane $w_i + u_i(x) = w_j + u_j(x)$ changes the assignment and the affine piece (check T8).*

(b) If $r_i = r_j$, the classical fibre contains the weight-split segment $\{w_i + w_j = c\}$. Its valuation image is the bent line $\{\min(w_i^{\mathbb{T}}, w_j^{\mathbb{T}}) = \operatorname{val} c\}$, made of two rays.

12.2 Statistical models as varieties

A parametric model is algebraic if $\theta \mapsto \mathbb{E}_\theta[s(X)]$ is polynomial or rational for a vector of sufficient statistics s. Its image is the *model variety*. For $N(\mu, \Sigma)$ on $\mathbb{R}^d$ with moments up to order r, the image is the Gaussian moment variety $G_{d,r}$, of dimension $d + d(d+1)/2$ when the parametrisation is generically injective. A mixture of k Gaussians has moment vector $\sum_i p_i m_r(\mu_i, \Sigma_i)$, so its image is the positive part of $\operatorname{Sec}_k(G_{d,r})$ [10]. Its parameter count is

$$\dim \Theta_{d,k} = k\big(d + \tfrac{d(d+1)}{2}\big) + k - 1,$$

and its expected dimension is $\min\{k \dim G_{d,r} + k - 1,\ \text{ambient dimension}\}$. The model is *generically identifiable* when the mixture map is generically finite-to-one, which requires $\dim \operatorname{Mix}_k = \dim \Theta_{d,k}$. It is *defective* when Sec_k has smaller than expected dimension. Defectivity is a global property of Sec_k, not a locus.

Proposition 12.3 (Moment order and the tower)**.** *(a) The order-r moment map of a two-component univariate Gaussian mixture has generic Jacobian rank* $\min(r, 5)$. *At order two the single-Gaussian image $\{(m, v + m^2)\}$ is already open, so second moments cannot see mixtures, and local identifiability needs $r \geq 5$ (check D1).*

(b) Affine statistics are constant on barycentre fibres. Regrouping a mixture of mixtures leaves the flattened law and all its moments unchanged (check D2). Hence the moment image of every tower level equals that of its flattening. If every node has m children, level n lies in Mix_{m^n}, *and the hierarchy itself is invisible to secant varieties. It lives in the convex-order fibre (proposition 5.2).*

Proof. (a) is direct computation at a rational point. (b) Moments are integrals, which are affine in the measure, and μ preserves integrals of affine functions. □

12.3 Walls of a component collision

For $\nu_\varepsilon = p\,N(\mu,\Sigma) + (1-p)\,N(\mu+\varepsilon v,\Sigma)$ with $0<p<1$, $\Sigma \succ 0$ and $v \neq 0$, the two-component fibre over ν_ε is finite for $\varepsilon \neq 0$, up to label swap. At $\varepsilon = 0$ it becomes the segment $\{(p_1,p_2;\mu,\Sigma) : p_1+p_2=1\}$. The collision locus is $\mathrm{Mix}_{k-1} \subset \mathrm{Mix}_k$, where the k-component parametrisation has positive-dimensional fibres. Tropically, by proposition 12.2(b), the collision is the locus $u_i = u_j$, where the tropical fibre gains a bent line.

Conjecture 12.4 (Wall-crossing, proposed)**.** *For a path ν_t crossing the collision locus transversally at ν_0, the change in the fibre is a finite sum $\sum_\sigma c_\sigma F_\sigma(\nu_0)$ over the collision strata σ through ν_0, with combinatorial coefficients c_σ determined by the local structure of the collision. In the single-collision example, $c_\sigma = 1$ and F_σ is the weight-split segment.*

The formal template is the wall-crossing of enumerative geometry, where the difference across a wall is a sum over strata. This is an analogy, not an established correspondence. The tropical picture of proposition 12.2 offers a concrete first test: the change in the tropical fibre across a tie or collision locus is a polyhedral computation.

Open problem 12.5 (Tropical identifiability)**.** For $\rho \in \mathrm{Mix}_k(V)$, is $\mathrm{val}(F_\rho^{(k)} \cap D_k(V)) = F^{(k)}_{\mathbb{T},\mathrm{val}\,\rho} \cap D_k^{\mathbb{T}}(\mathrm{trop}^+ V)$? For $V = \mathsf{D}_K X$ this is theorem 11.6. Lifting failures inside V would be tropical certificates of non-identifiability.

Open problem 12.6 (Order on tropical fibres)**.** Is the transitive closure of tropical pooling a partial order on $F_{\mathbb{T},v}$, with the two points of proposition 11.7 as bottom and top, and does it orient the tropical Beck defects of λ^σ as proposition 5.4 does classically?

Open problem 12.7 (Lifting obstructions as cohomology classes)**.** Sections 14 and 15 give two pieces of cohomology. One is the L^2 presheaf, whose Beck–Chevalley defect is measured by c. The other is the cohomology of tropical fibres, computed from assignment-cell nerves and determined by Farkas certificates. The Beck defects themselves are restrictions, not cocycles (proposition 14.7). The remaining question is whether lifting failures *inside a model* (open problem 12.5) define a coefficient system on the nerve whose H^1 is nonzero. That would make the tower's liftability certificates genuine cohomology classes.

13 The Hardy–Weinberg atlas

This section works one example through every layer of the paper. All the figures are produced by `atlas/atlas.py`, which computes each quantity from the formulas above; no data are sampled. The base is $X = \{0,1,2\}$, so a distribution is a point of a triangle. The model is the Hardy–Weinberg curve

$$r(\theta) = \big(\theta^2,\ 2\theta(1-\theta),\ (1-\theta)^2\big),$$

a positively parametrised toric curve, $r = (s^2, 2st, t^2)/(s+t)^2$. The tower uses three of its points, $\theta = 0.2, 0.5, 0.8$, with weights $\Pi = (0.35, 0.3, 0.35)$. An interactive version of the same figures accompanies the paper.

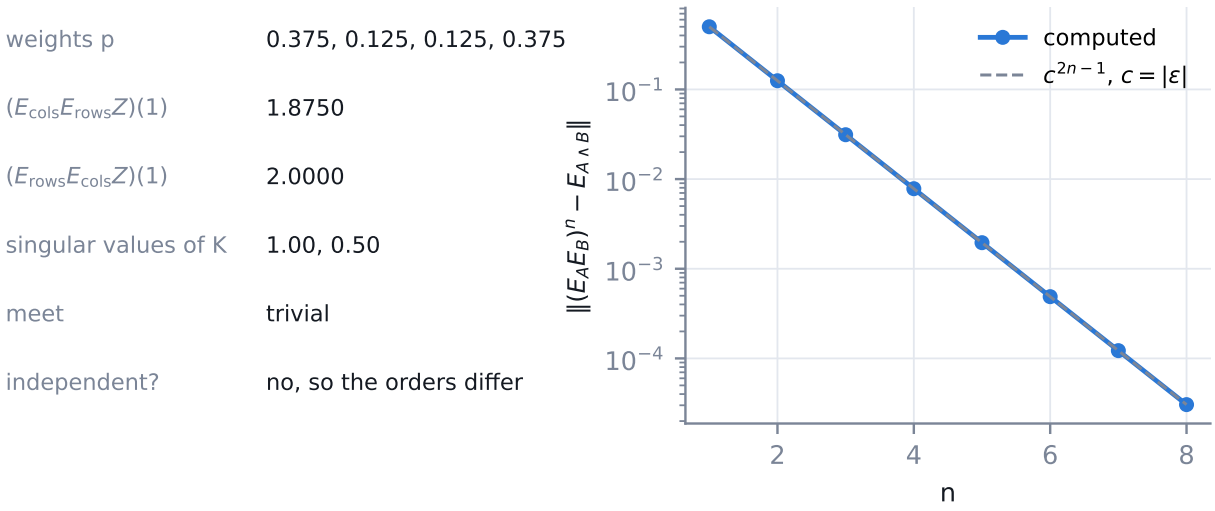


Figure 1: Algebra (theorems 3.3 and 3.6). Four points with weights $\frac{1\pm\varepsilon}{4}$, pooled by rows and by columns, at $\varepsilon = \frac{1}{2}$ (example 3.5). The two orders give 1.875 and 2.000 at point 1. The computed norms $\|(E_AE_B)^n - E_{A\wedge B}\|$ lie on c^{2n-1} with $c = \frac{1}{2}$, from 0.5 at $n = 1$ to 3.05×10^{-5} at $n = 8$.

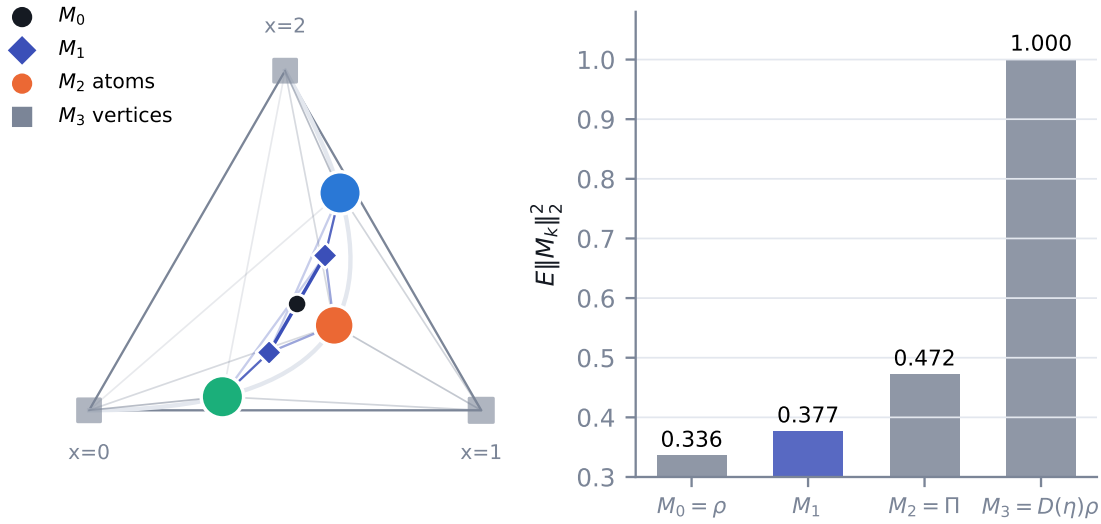


Figure 2: The tower as a martingale (proposition 5.2 and theorem 5.3). $M_0 = \rho$ (black), the level-two barycentres M_1 (diamonds), the atoms M_2 of Π (coloured, sized by mass) and the vertices M_3 (sized by ρ). Each position is a conditional expectation of the next. The energies $E\|M_k\|_2^2 = 0.336,\ 0.377,\ 0.472,\ 1.000$ increase along the convex-order chain. The outer pair, $\delta_\rho = e_\perp\Pi$ and $\mathsf{D}(\eta)\rho = e_\top\Pi$, are the two sides of the shift law's unit defect (proposition 5.4).

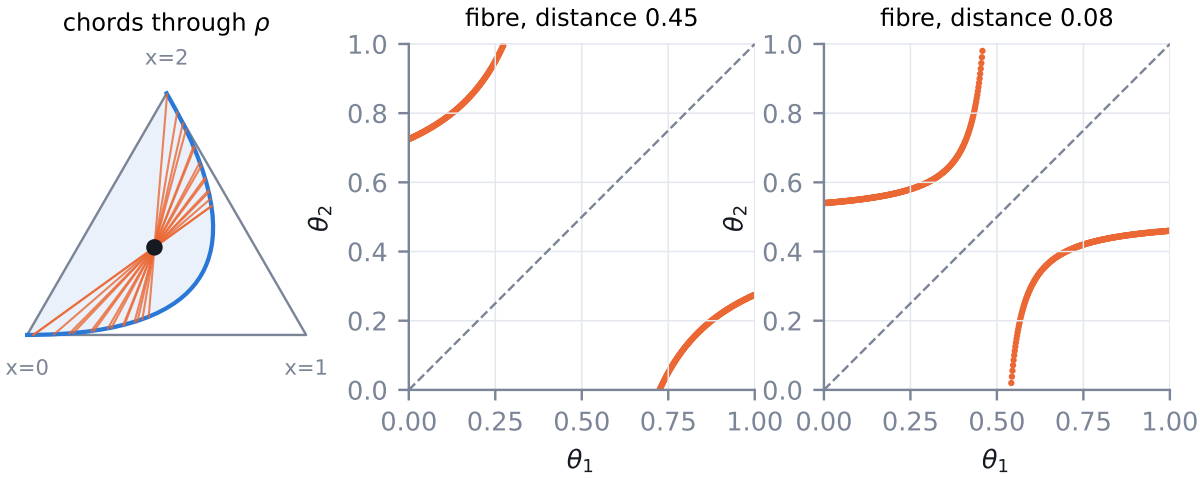


Figure 3: Secant geometry (section 12). Two-component mixtures of the curve fill its convex hull. The fibre over ρ is the family of chords through ρ, drawn as a curve in (θ_1, θ_2) coordinates. It has two branches, exchanged by relabelling the components. As ρ approaches the curve (distance $0.45 \to 0.08$), the branches close in on the diagonal $\theta_1 = \theta_2$, the collision locus where the fibre becomes the weight-split segment (proposition 12.2(b)).

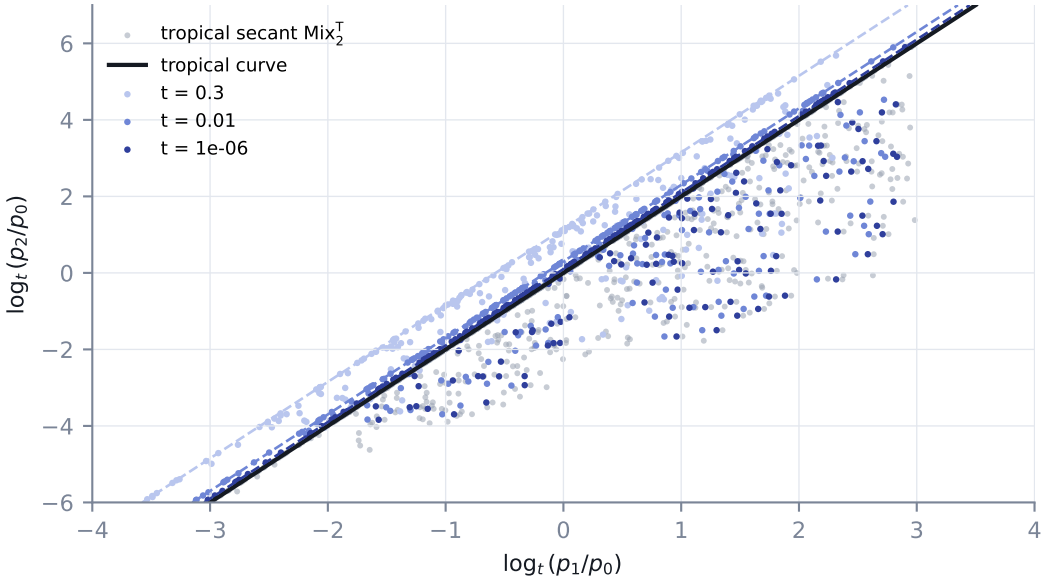


Figure 4: The Berkovich retraction in the chart $(\log_t(p_1/p_0), \log_t(p_2/p_0))$ (theorems 11.3, 11.9 and 11.11). Grey: the tropical secant $\min(w_1 + u_1, w_2 + u_2)$. Black: the tropical curve, which is the row space of the design. Coloured: the same 260 two-component mixtures at scales $t = 0.3$, 10^{-2}, 10^{-6}, and the image of the curve (dashed, shifted by $\log_t 2$). The largest coordinate deviation from the tropical limit is 1.60, 0.45 and 0.13. It decays like $1/\log(1/t)$, because only the leading-coefficient corrections $\log_t c$ remain.

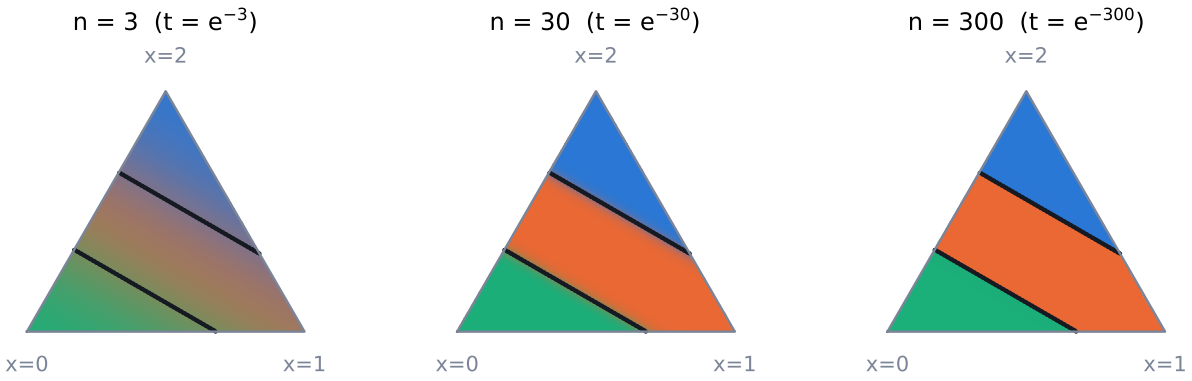


Figure 5: Zero-temperature shadows (propositions 12.1 and 12.2). Each empirical distribution q of n observations is coloured by the posterior over the three atoms, $\propto \Pi_i \exp(n \sum_x q(x) \log r_i(x))$. As n grows, which is $t = e^{-n} \to 0$, the colours harden into cells. The black walls are the $n \to \infty$ tie loci; they are straight because log-likelihoods are linear in q. The limiting map is nearest-atom pooling, discontinuous exactly on the walls (proposition 5.5).

14 Cohomology I: the L^2 presheaf and its Beck–Chevalley condition

Conditional expectations are the restriction maps of a presheaf, and the commutation theorem is its Beck–Chevalley condition. This section makes both statements precise and measures the failure.

Definition 14.1 (The L^2 presheaf)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $\mathcal{S}$ be the poset of sub-σ-algebras of $\mathcal{F}$ (completed), ordered by inclusion. The presheaf $\mathcal{H}$ on $\mathcal{S}$ is defined by

$$\mathcal{H}(G) = L^2(G), \qquad \rho_{G'G} = E_G|_{L^2(G')} \colon \mathcal{H}(G') \to \mathcal{H}(G) \quad (G \subseteq G').$$

Each restriction has a section, the inclusion $\iota_{GG'} \colon L^2(G) \hookrightarrow L^2(G')$, with $\rho_{G'G}\iota_{GG'} = \mathrm{id}$.

Lemma 14.2. *$\mathcal{H}$ is a presheaf: $\rho_{GG} = \mathrm{id}$ and $\rho_{G'G}\rho_{G''G'} = \rho_{G''G}$ for $G \subseteq G' \subseteq G''$.*

Proof. This is the tower property $E_G E_{G'} = E_G$. □

Definition 14.3 (Beck–Chevalley)**.** A commuting square of inclusions $C \subseteq A$, $C \subseteq B$, $A \subseteq D$, $B \subseteq D$ satisfies the *Beck–Chevalley condition* for $\mathcal{H}$ if

$$\rho_{DA} \circ \iota_{BD} = \iota_{CA} \circ \rho_{BC} \colon L^2(B) \to L^2(A),$$

that is, $E_A f = E_C f$ for every $f \in L^2(B)$. Its *defect* is the operator $\beta_{A,B} = (\rho_{DA}\iota_{BD} - \iota_{CA}\rho_{BC})$ on $L^2(B)$.

Theorem 14.4 (Beck–Chevalley is conditional independence)**.** *Let A, B be sub-σ-algebras (completed), $C = A \wedge B = A \cap B$, and $D \supseteq A \vee B$. The following are equivalent:*

(i) the Beck–Chevalley condition holds for the square $C \subseteq A, B \subseteq D$;

(ii) $E_A E_B = E_{A \wedge B}$ (condition (ii) of theorem 3.3);

(iii) $E_A E_B = E_B E_A$;

(iv) A and B are conditionally independent given $A \wedge B$.

In general the defect satisfies

$$\|\beta_{A,B}\| = \|E_A E_B - E_{A\wedge B}\| = c,$$

where c is the cosine of the Friedrichs angle between $L^2(A)$ and $L^2(B)$. For finite partitions, c is the largest singular value below 1 of the normalised contingency matrix K. The condition is symmetric in A and B.

Proof. Write $M = L^2(A) \cap L^2(B)$.

Step 0: $M = L^2(C)$. The inclusion $\supseteq$ is clear. Conversely, let f have an A-measurable version f_A and a B-measurable version f_B. For rational q, the sets $\{f_A > q\} \in A$ and $\{f_B > q\} \in B$ differ by a null set. Both σ-algebras are complete, so each set lies in both, and f_A is $A \cap B$-measurable.

Step 1: (i)⇔(ii). By definition, (i) says $E_A g = E_C g$ for all $g \in L^2(B)$, that is, $E_A E_B = E_C E_B$. Since $C \subseteq B$, the tower property gives $E_C E_B = E_C$.

Step 2: (ii)⇒(iii). E_C is self-adjoint, so $E_B E_A = (E_A E_B)^* = E_C^* = E_C = E_A E_B$.

Step 3: (iii)⇒(ii). Put $P = E_A E_B$. Commutation gives $P^2 = E_A^2 E_B^2 = P$ and $P^* = E_B E_A = P$, so P is an orthogonal projection. Its range lies in $L^2(A)$, and since $P = E_B E_A$ it also lies in $L^2(B)$, hence in M. Every $f \in M$ has $Pf = f$. So P is the projection onto $M = L^2(C)$, which is E_C.

Step 4: (ii)⇒(iv). Let $U \in L^\infty(A)$, $V \in L^\infty(B)$ and $W \in L^\infty(C)$. Because $UW \in L^2(A)$ and $E_A V = E_A E_B V = E_C V$,

$$\mathbb{E}[UVW] = \langle UW, E_A V\rangle = \langle UW, E_C V\rangle = \mathbb{E}[U\,W\,E_C V] = \mathbb{E}[E_C U \cdot E_C V \cdot W],$$

where the last step conditions on C, using that $W\,E_C V$ is C-measurable. Hence $\mathbb{E}[UV \mid C] = E_C U \cdot E_C V$.

Step 5: (iv)⇒(ii). For bounded $U \in L^2(A)$ and $V \in L^2(B)$, $\langle U, E_A V\rangle = \mathbb{E}[UV] = \mathbb{E}[E_C U\,E_C V] = \langle U, E_C V\rangle$. Bounded functions are dense in $L^2(A)$, so $E_A V = E_C V$. Both sides are continuous in V and bounded V are dense in $L^2(B)$, so $E_A E_B = E_C E_B = E_C$.

Step 6: the norm. Split $g \in L^2(B)$ as $g = g_M + g_\perp$ with $g_M \in M$ and $g_\perp \in L^2(B) \ominus M$. Then $\beta g_M = g_M - g_M = 0$ and $E_C g_\perp = 0$, so $\beta g = E_A g_\perp$. Moreover $E_A g_\perp \perp M$, since $\langle E_A g_\perp, m\rangle = \langle g_\perp, m\rangle = 0$ for $m \in M$. Hence

$$\|\beta\| = \sup\big\{|\langle f, g\rangle| :\ f \in L^2(A) \ominus M,\ g \in L^2(B) \ominus M,\ \|f\| = \|g\| = 1\big\},$$

which is the definition of the Friedrichs cosine [24]. Also $E_A E_B - E_C = (E_A - E_C)E_B = \beta E_B$, and E_B maps the unit ball of $L^2(D)$ onto that of $L^2(B)$, so $\|E_A E_B - E_C\| = \|\beta\|$.

Step 7: the finite formula. In the bases $1_a/\sqrt{p(a)}$ and $1_b/\sqrt{p(b)}$, $E_A|_{L^2(B)}$ has matrix K. Its singular values are the cosines of the principal angles, and the value 1 belongs exactly to M (lemma 3.1). Removing M leaves the largest remaining singular value. Symmetry follows from (iv). □

Remark 14.5 (Reading the theorem)**.** The lattice of σ-algebras, or of partitions, which are assignments of outcomes to cells, carries the presheaf $\sigma \mapsto L^2(\sigma)$. Its restriction maps are conditional expectations. Two restrictions interchange exactly when the square through the meet is Beck–Chevalley, and the size of the failure is one number, the Friedrichs cosine. The same number is the exact rate of alternating projections (theorem 3.6), so it measures both how far the square is from commuting and how fast iterated restriction forgets the difference.

Check H1 verifies the equivalence on 300 random squares (199 conditionally independent, 101 not), and H2 verifies $\|\beta\| = c$.

14.1 Gluing and the degree-one obstruction

For a cover $\{A, B\}$ of D with overlap $C = A \wedge B$, form the Čech-type complex

$$0 \to L^2(D) \xrightarrow{\delta^0} L^2(A) \oplus L^2(B) \xrightarrow{\delta^1} L^2(C) \to 0,$$

with $\delta^0 f = (E_A f, E_B f)$ and $\delta^1(f_A, f_B) = E_C f_A - E_C f_B$. A pair in $\ker \delta^1$ is *compatible*: it agrees on the overlap. A *gluing* of a compatible pair is an f with $\delta^0 f = (f_A, f_B)$. The degree-one cohomology is $\check{H}^1 = \ker \delta^1 / \operatorname{im} \delta^0$.

Theorem 14.6 (Gluing)**.** *(a) $\delta^1 \delta^0 = 0$, and δ^1 is surjective.*

(b) $\overline{\operatorname{im} \delta^0} = \ker \delta^1$. Hence $\check{H}^1 = 0$ iff $\operatorname{im} \delta^0$ is closed, iff $L^2(A) + L^2(B)$ is closed, iff $c < 1$. This always holds in finite dimensions, so every compatible pair glues.

(c) If $c = 1$, then $\check{H}^1$ is nonzero and non-Hausdorff: its reduced version $\ker \delta^1 / \overline{\operatorname{im} \delta^0}$ vanishes, but some compatible pairs do not glue.

(d) Under Beck–Chevalley the gluing is canonical: $f = f_A + f_B - E_C f_A$. This formula glues every compatible pair iff Beck–Chevalley holds.

(e) Gluings are not unique: $\ker \delta^0 = L^2(D) \ominus \big(L^2(A) + L^2(B)\big)$, which is nonzero unless $L^2(D) = \overline{L^2(A) + L^2(B)}$. So $\mathcal{H}$ satisfies the existence half of the sheaf condition (when $c < 1$), but not the uniqueness half.

Proof. (a) $E_C E_A = E_C = E_C E_B$, and $\delta^1(f, 0) = E_C f$ is onto $L^2(C)$. (b) The adjoint is $(\delta^0)^*(u, v) = u + v$, with kernel $\{(u, -u) : u \in L^2(A) \cap L^2(B) = L^2(C)\}$. So $\overline{\operatorname{im} \delta^0} = \{(f_A, f_B) : f_A - f_B \perp L^2(C)\} = \ker \delta^1$. By the closed range theorem, $\operatorname{im} \delta^0$ is closed iff $\operatorname{im}(\delta^0)^* = L^2(A) + L^2(B)$ is closed, which holds iff $c < 1$ [24]. (c) A dense, non-closed range is a proper subspace of its closure. (d) If Beck–Chevalley holds, $E_A f = f_A + E_C f_B - E_C f_A = f_A$, and symmetrically for B. Conversely, the pair $(f_A, E_C f_A)$ is compatible, and the formula returns $f = f_A$, which glues it iff $E_B f_A = E_C f_A$. Requiring this for all f_A is Beck–Chevalley. (e) $\ker \delta^0 = L^2(A)^\perp \cap L^2(B)^\perp$. □

Checks H3 and H4 verify (b) and (d) exactly on 300 squares.

Proposition 14.7 (Comparisons are restrictions, not gluing isomorphisms)**.** *Each Beck comparison computed in this paper (proposition 7.2) is a restriction $\rho_{G'G}$ with $G \subsetneq G'$. Such a map is surjective and not injective, and it strictly lowers the norm off $L^2(G)$. In particular no comparison map is an isomorphism, so the comparisons cannot be the gluing isomorphisms of a structure sheaf. What they are is the structure maps of $\mathcal{H}$. The interchange of two of them is the Beck–Chevalley condition (theorem 14.4), and its failure is measured by c.*

Proof. $\rho_{G'G} \iota_{GG'} = \mathrm{id}$ gives surjectivity, and $L^2(G') \ominus L^2(G) \neq 0$ is its kernel. Pythagoras gives the strict decrease (check H5). □

15 Cohomology II: fibres as topology

Classical barycentre fibres are convex (proposition 5.2), so their topology is trivial. So are the tower's level-two lifting fibres, which are polytopes [1, Def. 10.14]. Nontrivial topology appears in two places: when components are constrained to a curved model, and in tropical fibres, which are unions of polyhedra. This section computes the second kind exactly.

15.1 The cover by assignment cells

For the Hardy–Weinberg model, a tropical two-component decomposition has parameters $\vartheta = (\beta, d_1, d_2) \in \mathbb{R}^3$. Here β is the weight difference, $w_1 = \max(0, -\beta)$, $w_2 = \max(0, \beta)$, and d_i is the rate of component i, so that by theorem 11.9 $u(d) = (0, d, 2d) - \min(0, 2d)$. The *tropical fibre* over a target $v \in \mathsf{D}_{\mathbb{T}}X$ is

$$F_v = \big\{\vartheta : \min_i \big(w_i + u(d_i)(x)\big) = v(x) \text{ for all } x\big\}.$$

Definition 15.1 (Assignment cells)**.** For $\alpha\colon X \to \{1, 2\}$ (a designated attainer for each outcome) and a sign pattern $\sigma \in \{\pm\}^3$ of (β, d_1, d_2), let

$$P_{\alpha,\sigma} = \big\{\vartheta \in \text{orthant } \sigma : \; w_{\alpha(x)} + u(d_{\alpha(x)})(x) = v(x), \quad w_j + u(d_j)(x) \geq v(x) \; \forall j, x\big\}.$$

Lemma 15.2. *Each $P_{\alpha,\sigma}$ is a closed convex polyhedron. The cells cover F_v, and every intersection of cells is a polyhedron.*

Proof. Within an orthant every term is affine in ϑ, so each cell is defined by finitely many affine equalities and inequalities. A point of F_v lies in the cell given by its orthant and by any choice of attainer at each x. □

Theorem 15.3 (Fibre cohomology from the nerve)**.** *Let N_v be the nerve of the cover $\{P_{\alpha,\sigma}\}$. Then $H^*(F_v; \mathbb{Q}) \cong \check{H}^*(\{P_{\alpha,\sigma}\}; \mathbb{Q}) \cong H^*(N_v; \mathbb{Q})$. A set of cells spans a face of N_v iff its joint system of affine constraints is feasible. By Farkas' lemma, it is* not *a face iff there is a certificate $(y_{\mathrm{eq}}, y_{\mathrm{ub}} \geq 0)$ with $A_{\mathrm{eq}}^{\top} y_{\mathrm{eq}} + A_{\mathrm{ub}}^{\top} y_{\mathrm{ub}} = 0$ and $b_{\mathrm{eq}} \cdot y_{\mathrm{eq}} + b_{\mathrm{ub}} \cdot y_{\mathrm{ub}} < 0$. So the cohomology of the fibre is determined by which assignment patterns are certified impossible.*

Proof. The nerve theorem holds for finite covers by closed convex sets in $\mathbb{R}^d$: every nonempty intersection is convex, hence contractible [23]. The face criterion is Farkas' lemma [25]. □

These certificates have the same form as the tower's lifting certificates [1, Thm. 10.8, Thm. 10.13(iii)]. There, a separating functional certifies that a claim has no admissible lift. Here, a Farkas vector certifies that a pattern of attainers has no tropical decomposition. Theorem 15.3 upgrades such certificates from yes/no answers to cohomological data: the complex of patterns that are *not* certified impossible computes the homology of the fibre.

15.2 Local lifts as sections

Proposition 15.4 (Local lifts)**.** *Work in the full simplex model $\mathsf{D}_K X$, and fix a positive ρ with* $\mathrm{val}\,\rho = v$.

(a) For each cell, the construction in the proof of theorem 11.6, with the remainder at x given to $\alpha(x)$, is a section of val *over the rational points of the cell.*

(b) Over a fixed tropical point, the set of lifts is convex. A convex combination of two exact decompositions of ρ with the same valuation pattern is again one, because leading coefficients of positive terms add without cancelling.

(c) Consequently, on every nonempty face of the nerve, local lifts from different cells are joined by a segment of lifts. The lifting data add no obstruction beyond the emptiness certificates of theorem 15.3.

Proof. (a) is theorem 11.6 with designated attainers. (b) holds because the decomposition equations are linear in $(q_i(x))$, and the valuation pattern is fixed by positivity (check K4). (c) follows from (b) on each common point. □

For a model V, lifting into V is open problem 12.5. A lifting failure inside V would be a new kind of certificate. Its interaction with the nerve is the natural next computation.

15.3 The computation for Hardy–Weinberg

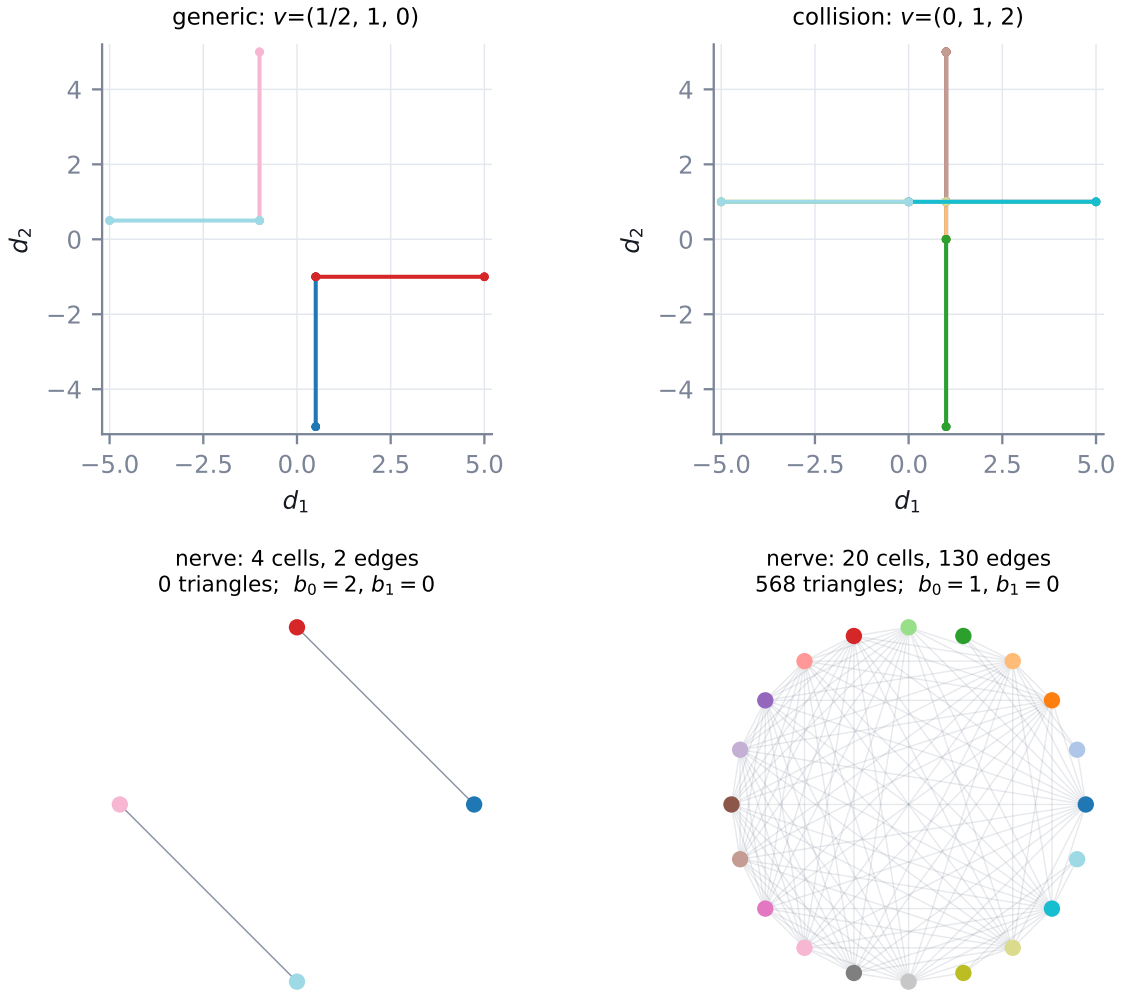


Figure 6: Tropical fibres F_v projected to (d_1, d_2), coloured by assignment cell, and their nerves. *Generic* $v = (\frac{1}{2}, 1, 0)$: four cells and two edges. There are two components, each a bent line, exchanged by swapping the components, and $b_0 = 2$, $b_1 = 0$. *Collision* $v = (0, 1, 2)$, on the tropical curve: twenty cells, 130 edges and 568 triangles. The branches are joined, and $b_0 = 1$, $b_1 = 0$.

Theorem 15.5 (Topology of the Hardy–Weinberg fibres)**.** *For the generic target* $v = (\frac{1}{2}, 1, 0)$*,* F_v *has Betti numbers* $(b_0, b_1) = (2, 0)$*. Its two components are exchanged by the label swap* $(\beta, d_1, d_2) \mapsto (-\beta, d_2, d_1)$*. For the collision target* $v = (0, 1, 2)$*, which lies on the tropical curve,* $(b_0, b_1) = (1, 0)$*.*

Proof. Exact computation (checks K1–K3). Every nonempty cell and face has an exact rational witness: 6 for the generic target and 718 for the collision target. Every empty cell and face among the candidates has an exact Farkas certificate: 64 and 104 respectively. The Betti numbers are ranks over $\mathbb{Q}$ of the coboundary matrices of the nerve. □

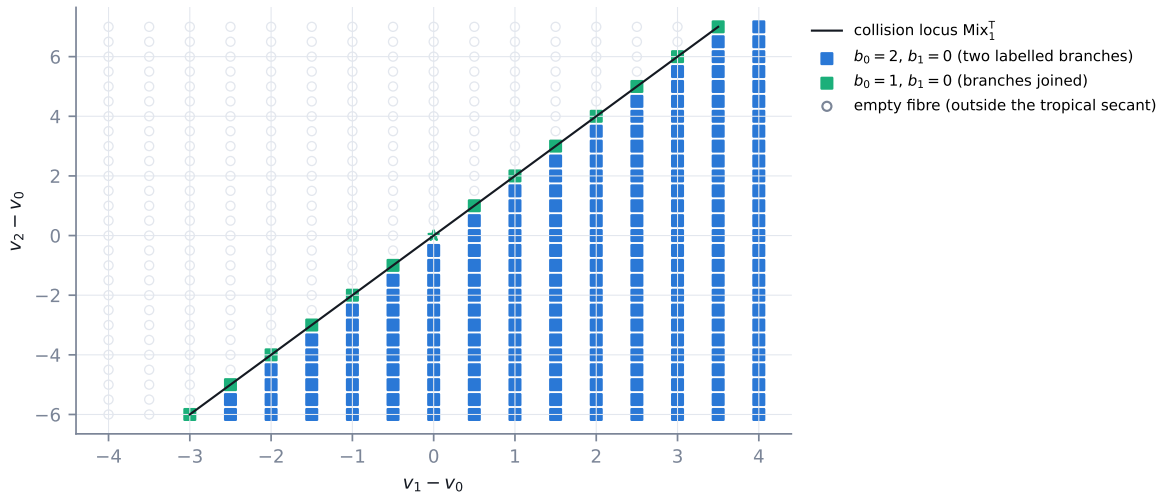


Figure 7: The homology of the fibre as a function of the target, over a grid of 17×27 targets in the chart $(v_1 - v_0, v_2 - v_0)$. Of these, 236 lie outside the tropical secant (empty fibre). On the 209 targets inside the secant and off the collision locus, $b_0 = 2$; on the 14 targets on the locus, $b_0 = 1$. $b_1 = 0$ wherever it was computed, which is every target except $v = (0, 0, 0)$, where all 64 cells are nonempty and only b_0 was computed. This is the homological form of the collision wall of section 12.

Reading. Over a generic target there are two continuous families of decompositions, one for each labelling. On the collision locus a component may coincide with the other, and the families merge. The same jump appears classically in fig. 3, where the two branches of the classical fibre meet the diagonal. Homology turns the proposed wall-crossing of conjecture 12.4 into a computable invariant. In this example the jump is $\Delta b_0 = -1$ across the collision locus, with no change in b_1. On the scanned grid, tie walls in the target (proposition 12.2(a)) change the cell structure of the fibre but not its homology.

16 Example: hallucination in a three-token model

This section connects the L^2 presheaf to the tower's notion of hallucination [1, §10.5]. A hallucination there is an *obstructed claim*: a statement that is admissible locally but has no coherent completion. We use the smallest model in which this happens, three token positions observed pairwise. Every number below is checked exactly by `check_hallucination.py`.

16.1 The model

Three binary token positions X_0, X_1, X_2 are seen through the contexts (windows) $\{0,1\}$, $\{1,2\}$ and $\{0,2\}$. In each context the two tokens agree with probability ε:

$$P_{X_iX_j} = \begin{pmatrix} \varepsilon/2 & (1-\varepsilon)/2 \\ (1-\varepsilon)/2 & \varepsilon/2 \end{pmatrix}.$$

The tables come from a mixture of sources $s \in \{A, B, U\}$, each with its own agreement ε_s. With weights w in the simplex, the pairwise tables mix linearly:

$$\varepsilon(w) = w_A\varepsilon_A + w_B\varepsilon_B + w_U\varepsilon_U.$$

We take $\varepsilon_A = 0.20$, $\varepsilon_B = 0.30$ and $\varepsilon_U = 0.40$.

16.2 Hallucination is a failure of gluing

Proposition 16.1 (Global sections and their certificate)**.** *(a) A joint law of (X_0, X_1, X_2) with these pairwise tables exists iff $\varepsilon \geq \frac{1}{3}$.*

(b) For $\varepsilon < \frac{1}{3}$, the functional $\Phi(P) = \sum_{i<j} P(X_i \neq X_j)$ certifies the obstruction. Every joint has $\Phi \leq 2$, because each of the eight joint outcomes has at most two disagreeing pairs. The pairwise tables give $\Phi = 3(1-\varepsilon)$, which exceeds 2 by exactly $1 - 3\varepsilon$.

(c) For $\varepsilon \geq \frac{1}{3}$, the global sections form the segment $J + s\chi$, $|s| \leq \min(a, b)$. Here $\chi(x) = (-1)^{x_0+x_1+x_2}$ spans the one-dimensional kernel of the pairwise-margin map, and J is the symmetric joint with $a = \frac{3\varepsilon-1}{4}$ on 000 and 111 and $b = \frac{1-\varepsilon}{4}$ on the other six outcomes.

Proof. (a) and (b): this is [2, Thm. 4.8] at $\tau = 0$. All three correlations are $E_{ij} = 2\varepsilon - 1$, and the 3-cycle inequality with all signs negative reads $-3(2\varepsilon - 1) \leq 1$. The bound $\Phi \leq 2$ is checked atom by atom. (c) The margin map has rank 7 on $\mathbb{R}^8$, and χ lies in its kernel. J reproduces the tables. Nonnegativity of $J + s\chi$ gives the range of s (checks L1, L2). □

In the language of [1, §10], each pairwise table is an admissible local claim, and for $\varepsilon < \frac{1}{3}$ the family is an obstructed claim. Its certificate is the test function Φ, a separating functional of the kind in [1, Thm. 10.8]. The parity χ is not a certificate. It indexes the non-uniqueness of the joint when one exists.

16.3 The wall in weight space, and a token stream

By proposition 16.1, the mixture is liftable iff $\varepsilon(w) \geq \frac{1}{3}$. This is a half-space in the weight simplex: the wall $\varepsilon(w) = \frac{1}{3}$ is a straight line, and the liftable side is the side of the coherent source U (the only source with $\varepsilon_s \geq \frac{1}{3}$). In the coordinates $\theta = w_A/(w_A + w_B)$ and $\mu = w_U$, the wall becomes the curve $\mu(\theta) = (\frac{1}{3} + \theta)/(1 + \theta)$. The liftable side is $\mu \geq \mu(\theta)$: more weight on the coherent source restores a joint.

For the 12-step stream $(\theta_t, \mu_t) = (0.10, 0.10), (0.20, 0.15), \ldots, (0.30, 0.90)$ of fig. 8, the agreement ε_t rises from 0.301 to 0.387. The stream is obstructed (hallucinating) for $t = 1, \ldots, 7$, with $\varepsilon_7 = 0.332$, and liftable from $t = 8$, with $\varepsilon_8 = 0.337$. The transition is the crossing of the wall between steps 7 and 8 (check L6). This wall is classical: it is linear in the weights, and it is not a tropical wall (section 16.6).

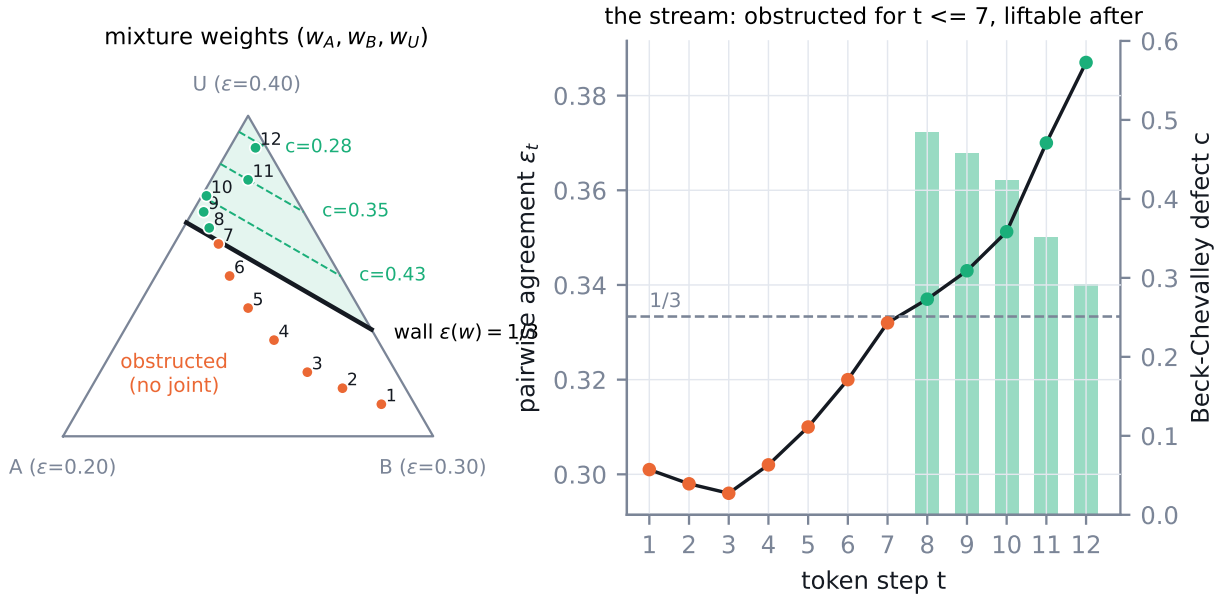


Figure 8: The three-token hallucination example. Left: the weight simplex. The straight wall $\varepsilon(w) = \frac{1}{3}$ separates the obstructed region (no joint law) from the liftable region (shaded). Dashed lines are level sets of the Beck–Chevalley defect c of the symmetric joint, and points are the 12 stream steps (orange obstructed, green liftable). Right: ε_t along the stream, which crosses $\frac{1}{3}$ between $t = 7$ and $t = 8$, and the defect c_t (bars) once a joint exists.

16.4 The L^2 presheaf on context windows

Once the claim is liftable, choose a full-support global section P (for instance J, for $\frac{1}{3} < \varepsilon < 1$). This gives a probability space, and the windows give two σ-algebras on it:

$$A = \sigma(X_0, X_1), \qquad B = \sigma(X_1, X_2).$$

Their shadows E_A and E_B are what a reader confined to one window can compute.

Proposition 16.2 (Window shadows)**.** *For a full-support section P:*

(a) the meet is $A \wedge B = \sigma(X_1)$, the shared token;

(b) the window square satisfies Beck–Chevalley iff $X_0 \perp X_2 \mid X_1$ under P, that is, iff the sequence is Markov through the shared token;

(c) the defect is $c = \max_{x_1} \left|\operatorname{corr}_P(X_0, X_2 \mid X_1 = x_1)\right|$, and alternating the two window shadows converges to $E[\cdot \mid X_1]$ at rate c^{2n-1};

(d) at the symmetric section J, $c = \dfrac{|2\varepsilon - 1|}{2\varepsilon}$;

(e) a Markov section exists in the segment $J + s\chi$ iff $\varepsilon = \frac{1}{2}$ (for $\frac{1}{3} < \varepsilon < 1$). On 99 sections across the segment, the defect is smallest at the symmetric section.

Proof. (a) The cells of A are values of (x_0, x_1) and those of B values of (x_1, x_2). Two cells meet iff they share x_1, so the components of $G(A, B)$ are the two values of x_1 (lemma 3.1). (b) This

is theorem 14.4(iv) with meet $\sigma(X_1)$. (c) K is block diagonal over the two components. Each block is the normalised 2×2 table of (X_0, X_2) given $X_1 = x_1$, whose singular values are 1 and the absolute correlation. Now apply theorems 3.6 and 14.4. (d) Given $X_1 = 0$, the table of (X_0, X_2) is proportional to $\left(\begin{smallmatrix} a & b \\ b & b \end{smallmatrix}\right)$. Its correlation is $(a-b)/(2(a+b)) = (2\varepsilon - 1)/(2\varepsilon)$, and the case $X_1 = 1$ is symmetric. (e) Under the Markov property, $P(X_0 = X_2 \mid X_1) = \varepsilon^2 + (1-\varepsilon)^2$. This must equal the table value ε, so $\varepsilon \in \{\frac{1}{2}, 1\}$. The minimisation is computed (checks L3–L5, L7). □

Reading. Two context windows that share a token commute as conditional expectations exactly when everything the first token says about the third passes through the shared token. The defect c is the long-range coupling that no single window represents, and c^{2n-1} is how fast repeated window-by-window conditioning forgets it. Along the stream, once a joint exists the defect falls from $c_8 = 0.484$ to $c_{12} = 0.292$ as weight moves to the coherent source. It never reaches 0, because $\varepsilon = \frac{1}{2}$ is not attainable with these sources.

16.5 Three layers of obstruction

regime	structure	certificate or measure
$\varepsilon < \frac{1}{3}$	no global section (hallucination)	$\Phi > 2$, by $1 - 3\varepsilon$ (proposition 16.1)
$\frac{1}{3} \le \varepsilon < 1$, $\varepsilon \neq \frac{1}{2}$	sections form a segment; windows do not commute	Beck–Chevalley defect $c > 0$ (proposition 16.2)
$\varepsilon = \frac{1}{2}$	a Markov section exists	$c = 0$ at that section

The first layer is a failure of gluing for the presheaf of distributions on the context cover. The second is the Beck–Chevalley defect of the L^2 presheaf on the glued space. The tower's obstructed claims live in the first layer. The L^2 presheaf measures what remains once the claim is coherent.

16.6 What tropicalisation sees

Proposition 16.3 (Tropical shadow of hallucination)**.** *Over a real non-Archimedean field:*

(a) valuation commutes with marginalisation, so if the tropical tables have no tropical joint, the classical tables have no joint;

(b) the hallucination at constant $\varepsilon < \frac{1}{3}$ is invisible tropically: all entries have valuation 0, and the zero tropical joint exists;

(c) a scaled family with agreement $\varepsilon = \kappa t^\alpha$, $\alpha > 0$, is tropically obstructed. The certificate is combinatorial: every outcome has an agreeing pair, so every tropical joint entry is at least α, while the disagreement cells require 0.

Proof. (a) Marginalisation is pushforward, and theorem 11.3 gives $\mathrm{val}(\mathsf{D}f(P)) = \mathsf{D}_{\mathbb{T}} f(\mathrm{val}\, P)$. (b) and (c) are direct (check L8). □

So the tropical shadow of a hallucination certificate is a support-level certificate: strong contextuality in the sense of [4], graded by rates. It is sufficient but not necessary. The boundary $\varepsilon = \frac{1}{3}$ concerns leading coefficients, which the valuation discards. Hallucination is therefore *not* in general a tropical wall crossing. The wall that detects it is classical and linear in the weights (fig. 8). The tropical walls of proposition 12.2 detect only obstructions that are visible in rates.

17 Empirical tests of the hallucination example

The tests were pre-registered before any of the statistics below was computed, in the file `PREREG_hallucination.md` of the folder `halluc_study` (its SHA-256 begins `4a34e1bb`). The same file discloses an earlier, different analysis of the same data in the validation paper. There were three studies. Study 1 runs on public data. Study 2 was run on samples from a local model; its analysis follows a frozen amendment. Study 3 needs pairwise elicitation from a model and ships as a kit with a synthetic smoke test.

17.1 Why only the second layer can be tested on existing data

The obstruction of proposition 16.1 needs each pair of tokens to be observed in a separate context. In existing benchmarks every item's answers are observed jointly, and a jointly observed table always has a joint law, so the gap below $\frac{1}{3}$ cannot appear in them. Study 3 is designed to collect pairwise data. On jointly observed data the testable layer is the second one: the window Beck–Chevalley defect of proposition 16.2.

17.2 Study 1: LLM-drift answers

Data. The data are the answers of GPT-3.5 (0301, 0613) and GPT-4 (0314, 0613) to the same items [5]: four runs per item, one answer each. Five tasks have a reference answer comparable by equality: USMLE (intact prompts), primality, happy numbers, HotpotQA (agent) and ARC. The extraction rules are those of the validation paper, unchanged.

P1: consistency detector. For a target run, the score is the fraction of the other runs whose valid answer differs from the target's. The outcome is whether the target answer is wrong. This is the cross-sample consistency principle of SelfCheckGPT [9], with the other runs playing the role of samples.

P2: window defect. The binary correctness of GPT-4 March, GPT-4 June and GPT-3.5 June is read through two windows:

$$A = \sigma(Y_{4\mathrm{M}}, Y_{4\mathrm{J}}) \text{ (same model, two dates)}, \qquad B = \sigma(Y_{4\mathrm{J}}, Y_{35\mathrm{J}}) \text{ (same date, two models)}.$$

They share $Y_{4\mathrm{J}}$. By proposition 16.2, Beck–Chevalley holds iff the March GPT-4 errors and the June GPT-3.5 errors are independent given the June GPT-4 error. The defect c is tested against a parametric Markov null.

Table 1: Study 1 (pre-registered). AUROC of the disagreement score for target runs GPT-4 June (primary) and GPT-3.5 June (secondary), and the window Beck–Chevalley defect c with its Markov-null median and Holm-adjusted p-value ($B = 2000$; 95% bootstrap intervals).

task	n	wrong	AUROC (4J)	AUROC (35J)	n_c	c	null	p_{Holm}
USMLE	340	52	0.692 [0.617, 0.766]	0.895 [0.853, 0.930]	338	0.096 [0.033, 0.401]	0.106	0.756
primes	1000	489	0.881 [0.861, 0.901]	0.739 [0.704, 0.771]	949	0.498 [0.411, 0.584]	0.047	0.003
happy numbers	500	324	0.873 [0.840, 0.902]	0.888 [0.855, 0.917]	387	0.098 [0.020, 0.211]	0.082	0.756
HotpotQA	4844	2693	0.708 [0.694, 0.721]	0.800 [0.783, 0.817]	324	0.309 [0.128, 0.519]	0.083	0.005
ARC	467	341	0.821 [0.779, 0.862]	0.902 [0.854, 0.945]	467	0.320 [0.200, 0.446]	0.069	0.003

Results.

- *H1 is supported.* Disagreement with the other runs detects wrong GPT-4 June answers in all five tasks, with AUROC between 0.69 and 0.88 and every interval above 0.5.
- *H2 is supported.* Beck–Chevalley fails in three tasks: primes ($c = 0.50$ against a null median of 0.05), HotpotQA (0.31) and ARC (0.32). In these tasks the errors of the two outer runs stay coupled after conditioning on the shared run. By proposition 16.2, this is coupling that neither window holds: a shared item-level difficulty that a single run does not screen off. In USMLE and happy numbers the defect is indistinguishable from the Markov null.
- *Exploratory.* The rank correlation between c and the GPT-4 June AUROC across the five tasks is 0.70. With five tasks this carries no inferential weight.

What Study 1 does and does not show. It shows that the window defect is measurable on real model outputs, and that it is nonzero in several tasks. It also replicates the consistency principle that underlies black-box hallucination detection. It does not test the obstruction below $\frac{1}{3}$, for the reason given above. Its errors are factual errors against a reference, a common proxy for hallucination [6] but not the same as an obstructed claim in the sense of [1, §10.5].

17.3 Study 2: HaluEval QA

Data and amendment. HaluEval QA [7] (MIT licence) gives, for each item, knowledge, a question, a right answer and a hallucinated answer. Llama 3.1 (8B, run locally through Ollama) was sampled $K = 5$ times per prompt at temperature 1 on items 0–1999, with the system line "Answer with a short phrase only". Items 0–199 were inspected first; that inspection found that answer length alone separates the two answer types, and that the containment judge misses paraphrases. Amendment 1 (`AMENDMENT1.md`, SHA-256 beginning `07782f4e`) was frozen after that and before any statistic on items 200–1999 was computed. It adds a length baseline, a length-adjusted AUROC, a logistic test of the detector beyond length, a relaxed deterministic judge, and a within-sample window test. Deviations from the original registration: $N = 2000$ rather than 10 000, the system line, and the choice of model. The confirmatory set is items 200–1999 ($n = 1800$).

Table 2: Study 2, confirmatory set ($n = 1800$ items, 3600 answers; Amendment 1). Detector = fraction of the five samples that do not match the answer. Length-adjusted AUROC compares answers only within the same length quintile. LR = likelihood-ratio statistic for adding the detector to log(length) in a logistic model. 95% bootstrap intervals over items ($B = 2000$).

judge	AUROC	length-adjusted AUROC	LR (p)
containment (pre-registered)	0.871 [0.860, 0.881]	0.851 [0.827, 0.874]	404 ($< 10^{-80}$)
relaxed (Amendment 1)	0.870 [0.859, 0.882]	0.883 [0.861, 0.903]	493 ($< 10^{-100}$)
length alone	0.974	—	—

Results.

- *The detector works, but length is the stronger cue in this benchmark.* The pre-registered AUROC is 0.871. Answer length alone reaches 0.974: HaluEval's hallucinated answers are usually longer than the right ones. So the raw AUROC cannot be read as evidence for the detector.

- *H1″ is supported: the detector adds information beyond length.* Within length quintiles the detector still separates the two answer types (AUROC 0.85 with the containment judge, 0.88 with the relaxed judge), and adding it to a length-only logistic model improves the fit by LR = 404 (cluster-robust Wald $p < 10^{-80}$, allowing for the two answers per item). The adjustment is uneven: the middle quintile holds most of the comparisons, and in the outer quintiles one answer type has only 7 to 31 members.

- *H2′ is supported: the within-sample windows fail Beck–Chevalley.* Let $Y_k = 1$ if sample k misses the right answer (relaxed judge; 14.7% of samples). With windows $\sigma(Y_1, Y_2)$ and $\sigma(Y_2, Y_3)$, the mass-weighted defect is $\bar{c} = 0.32$ against a permutation-null median of 0.03 ($p = 0.0005$). Adding the difficulty count $D = Y_4 + Y_5$ to the centre reduces it to 0.13 (null 0.02, $p = 0.005$).

Interpretation. The five samples are independent given the item, so the defect does not come from the sampling. It comes from a variable missing from the lattice: the item itself. The wrong-count distribution shows this. 1441 items have no wrong sample and 183 have all five wrong, against 813 and 0 expected under one common error rate. A model that is wrong on an item tends to be wrong in the same way every time. One sample in the centre does not screen this off; adding two more absorbs most of it, as the centre-enlargement reading of proposition 16.2 predicts. This is a common cause, not an obstruction: the samples are jointly observed, so a joint law exists. The 183 consistently wrong items are where a consistency detector is blind.

Does the test detect Beck–Chevalley correctly? The theorem side is exact: Beck–Chevalley holds iff the outer variables are independent given the centre (theorem 14.4, checks H1, H2, L4). The finite-sample test was checked separately on synthetic data (`check_bc_test.py`).

- *When Beck–Chevalley holds it rejects at the nominal rate.* On random Markov chains $Y_1 - Y_2 - Y_3$ the rejection rate at level 0.05 is 0.056 ($n = 300$) and 0.060 ($n = 1800$) over 1000 replications, with uniform p-values. The population defect is 0 for all of 200 random Markov laws.

- *When it fails it detects it.* With item-level latent difficulty (samples i.i.d. given the item) the population defect is 0.35. The power is 0.76 at $n = 100$ and 1.00 from $n = 300$, and the sample $\bar{c}$ matches the population value at every n.

- *Centre enlargement works as proposition 16.2 says.* With the latent class in the centre, Beck–Chevalley holds again and the rejection rate is 0.042. The proxy $D = Y_4 + Y_5$ only shrinks the defect ($0.35 \to 0.10$).

- *Replica of Study 2.* A three-class binomial model fitted to the wrong-count distribution alone gives simulated $\bar{c}$ intervals $[0.30, 0.56]$ (centre Y_2) and $[0.07, 0.41]$ (centre (Y_2, D)). Both contain the observed values 0.32 and 0.13. This model has as many parameters as free cells, so the replica shows consistency, not a test of the latent reading.

Limitations. Both judges are string-based. The relaxed judge still misses name variants ("Don Henley" against "Donald Hugh Henley"), which inflates the detector score of some right answers. A meaning-based judge is the next step. HaluEval's hallucinated answers were written to a template, which produces the length gap; the length-adjusted results are the ones to cite.

17.4 Study 3: kit

- *Study 3* (`kit_pairwise_cycles.py`) asks about each pair of three related claims in a separate prompt, and evaluates the 3-cycle inequalities of [2, Thm. 4.8]. It reports the largest violation V with a bootstrap interval. For the symmetric model $V = 2(1-3\varepsilon)$; the smoke test recovers $V \approx 0.5$ at $\varepsilon = 0.25$ and no violation at $\varepsilon = 0.45$. This is the design in which a hallucination in the sense of proposition 16.1 can be observed directly.

TruthfulQA [8] can supply claim triples for Study 3 once model answers are collected.

18 Computational checks

18.1 The scripts

- `check_shadows.py` has 30 checks, in exact rational arithmetic except A1(v) and A4, which use double precision with tolerance 10^{-9}. It imports the tower objects from `check_tower.py`.
- `check_tropical.py` has 8 checks, in exact Puiseux arithmetic except the floating-point ranks in T6 and T7.
- `check_cohomology.py` has 10 checks, exact except H2.
- `check_hallucination.py` has 9 checks, exact except L4b, L5 and L7.
- `check_bc_test.py` has 7 Monte Carlo checks (S1–S6) of the finite-sample Beck–Chevalley test of section 17.
- In the folder `halluc_study`, `study1_llmdrift.py` runs the pre-registered Study 1 (table 1), and `kit_halueval.py` and `kit_pairwise_cycles.py` are the kits for Studies 2 and 3, with smoke tests.
- `atlas/atlas.py` draws figs. 1 to 8, and `atlas/fibre_scan.py` computes the Betti map. Both use `atlas/fibre_topology.py`.

All checks pass; the hashes are in `FROZEN.txt`.

A1–A7b	theorem 3.3 (449 commuting, 151 non-commuting); proposition 3.4; example 3.5; theorem 3.6; corollary 3.8; lemma 3.2; proposition 3.9
B1–B7	star space; theorem 4.9; proposition 4.10; theorem 5.3; proposition 4.11; propositions 5.1 and 5.2; proposition 5.4
C1–C7	audit of earlier scripts; proposition 4.2; example 4.3; proposition 5.5(b)
D1–D2, E1–E2	proposition 12.3; theorem 9.4
G1–G3	proposition 10.1; proposition 4.6; proposition 6.8
T1–T8	theorem 11.3; theorem 11.6; theorem 11.11(a); proposition 12.1; example 11.10; proposition 12.2(a)
H1–H5	theorem 14.4; theorem 14.6(b, d); proposition 14.7
K1–K5	theorems 15.3 and 15.5; proposition 15.4(b); fig. 7
L1–L8	propositions 16.1 to 16.3; the token stream of fig. 8
S1–S6	the permutation test of section 17: size, power, centre enlargement, replica of table 2

18.2 What checks establish

Finite exact checks verify identities on the enumerated inputs and can falsify overbroad claims. They do not prove identities for all measures, establish measurability, or supply categorical typing. Every general statement in this paper has a proof; the checks guard the proofs against slips. Hashes certify the identity of artefacts, not the truth of theorems.

18.3 Audit of earlier scripts

Five scripts accompanied an earlier version. All five share one nearest-atom `pooling(nu, atoms)`, with "type-aware" total variation and ties broken by key order. They pass every reported check for the following reasons:

Script	Why it passes
`check_giry.py`	Π always has two atoms, and Σ's descriptors i mod 2 biject with them. So $\sigma(\theta) = \sigma(\tilde{r})$, and L and R have identical cells (C1).
`check_3cat-Coherence.py`	As above. In addition, level-5 atoms of type D^2 against inputs of type D^1 are all at TV distance 1, so the pooling is total, $e_\perp$, and the "3-cell" is trivial (C7).
`check_non-coherent.py`	The duplicated predictive $(0, \rho_1), (1, \rho_1)$ ties, and the first key wins, so R's cells equal L's (C6).
`double_category.py`	"Interchange" is $R(L\nu) = L(R\nu)$ with $L = R$. "(U) defect nonzero" is $e_\perp \neq e_\top$.
`check_span.py`	The cells coincide (C1). `dist_of` strips θ, so "not inverse" holds by a change of key type. "(Cm) equals sum" compares a pair with itself (C5). Mixed inputs combine two key types under one metric.

By proposition 4.2, any composite of nearest-atom poolings is one pooling onto a coarsening of the first partition. By proposition 12.1, such pooling is a zero-temperature shadow. The scripts' counts therefore carry no information about theorem 3.3. A meaningful test uses the star space with labels that are not functions of the predictive, and compares three pairs: canonical shadows, label-only shadows, and naive composites. Checks B2, B3 and C2 do exactly this.

19 Consequence for the tower paper

The tower paper [1] never uses $E_B E_A = E_{A \vee B}$. Its poolings live on a fixed sample index [1, Def. 13.13], and its order in [1, Thm. 13.16] is a transitive closure. An unnumbered paragraph added after that theorem explains why the closure is needed. A two-step label pooling $E_{G_2} E_{G_1}$ is one conditional expectation iff the two commute, which by theorem 3.3 means conditional independence given $G_1 \wedge G_2$; the product is then $E_{G_1 \wedge G_2}$, never the join.

20 Summary and open problems

- Two conditional expectations commute iff they are conditionally independent given their meet. Otherwise they interchange laxly, through the meet, at the exact rate c^{2n-1}. The join identity is false except for equal σ-algebras.

- The canonical shadows of a context live on its star space. They form a commutative band, are natural along context maps that are injective on predictive atoms, and interchange strictly.
- The tower acts on barycentre fibres as a martingale. $e_\perp$ and $e_\top$ are the bottom and top of each fibre. Each computed Beck defect is one conditional-expectation step.
- Over a real non-Archimedean field the whole picture tropicalises exactly: the valuation is a monad morphism, fibres lift, mixtures become tropical secants, and the valuation factors through Berkovich spaces. Hard assignment is the zero-temperature shadow.
- Conditional expectation is restriction in the L^2 presheaf. Commutation is its Beck–Chevalley condition, and gluing fails only when $c = 1$. Tropical fibres have computable cohomology, read off from certificates of impossible assignment patterns. In the Hardy–Weinberg example the collision wall is a jump $\Delta b_0 = -1$.
- In a three-token model a hallucination is a failure to glue pairwise windows, certified by a Bell-type functional, below a straight wall in source weights. Above the wall, overlapping windows commute iff the tokens are Markov through the shared token. The Friedrichs cosine is the exact measure of the coupling that no window holds.

The open problems are open problems 6.11, 8.1 and 12.5 to 12.7, and conjecture 12.4.

A Changes from the previous versions

This paper supersedes three manuscripts of October 2026:

(i) *Conditional Expectations, Beck Defects, and a Proposed Double-Category Structure*, referred to below as G;

(ii) *Conditional Expectations, Beck Defects, and the Shadow Algebra of the Posterior Tower*, referred to below as S;

(iii) *From Barycentre Fibres to Secant Geometry.*

S and (iii) are absorbed in full. The table lists every statement of G that changed.

G	Status	Here
Def 3.2, Props 3.3, 3.5 (label shadows)	kept	proposition 4.1; composites in proposition 4.2
Thm 4.2 (nested)	kept	special case of theorem 3.3
Thm 4.4 (CI ⇒ commute)	strengthened	five-way equivalence, theorem 3.3
Rem 4.5 (join), Ex 4.6	kept, sharpened	proposition 3.4; example 3.5
Rem 4.7 ("hypotheses necessary")	made precise	theorem 3.3; general case theorem 3.6 and corollary 3.8
Prop 4.8, Rem 4.9, Prop 8.1 (inheritance)	discharged	proposition 4.5
Defs 5.1, 5.3, Rem 5.4, Defs 5.5, 5.6, Props 5.7, 5.8, Rem 5.9	kept	section 6; context maps defined concretely in definition 6.1

G	Status	Here
Def 5.10, Prop 5.11 (compatible subcategory)	proved	propositions 4.6 and 6.8
Diagram after Def 5.13	withdrawn	horizontal composition of contexts is not a functor $\mathbf{Ctx} \times \mathbf{Ctx} \to \mathbf{Ctx}$
Thm 5.14 (interchange blueprint)	replaced	proposition 6.10
Prop 6.1 (required data)	kept	proposition 7.1, now supplied by proposition 7.2
Conj 6.2 (two-shadow factorisation)	superseded	one step suffices in every computed case
Thms 7.1, 7.2 (Options A, B)	partly invalid	proposition 7.4: (Cm) does not decompose
§8 (Giry)	completed	section 8
Prop 9.1, Rem 9.2 (3-cells)	kept	triviality in quintets
§10 (regimes)	kept	worked regimes in section 3
Def 11.1, Thms 11.2, 11.5 (double category of the tower)	withdrawn	2-cells mistyped; proposition 6.10
Thm 11.4, Prop 11.8	kept	identities and latent relabellings are natural (section 10)
Prop 11.7 (Gaussian interchange)	false	proposition 3.9
§11.6.1 (restricted domain V)	corrected	the square is total (section 10)
§12 (scope of checks)	kept	section 18
§13 (Berkovich programme)	replaced	section 11
§14 (stochastic-to-convex bridge)	kept, strengthened	theorem 9.4: fully faithful; shadows not affine
§15 (Wasserstein–Berkovich functor)	withdrawn	ill-defined; theorem 11.11 instead
§16 (secants, wall-crossing)	corrected	section 12: order ≥ 3; hierarchy invisible to moments; conjecture 12.4

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- S. Lin, J. Hilton and O. Evans for TruthfulQA [8];
- the developers of Python, NumPy, SciPy (including the HiGHS solver), SymPy, pandas and Matplotlib, on which every computation and figure depends.

The data are used under their licences and terms; no credentialed or patient-level data were used. The study in section 17 was pre-registered and frozen by hash before its statistics were computed. Independent reviews by language-model referees found errors in earlier drafts of this material, which are corrected here, and an AI assistant (Claude, Anthropic) helped draft text, write the exact-arithmetic checks and produce the figures. The author is responsible for all remaining errors.

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